Higher Harmonic Reconstruction and Exact Collision Identities

Gap ten, three-cluster polylogarithms,
and Gamma–Stieltjes primitives

Research continuation prepared for
Vladimir Reshetnikov's *ProveIt* project

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Abstract

We continue several explicitly open directions in the inspected ProveIt manuscript and incoming research collection. First, we evaluate the local correction for collisions of arbitrarily many polygamma factors along arbitrary positive paths. For nonresonant monomial scales this becomes a finite zeta polynomial. With one trigamma factor and any number of centered digammas, the complete constant is given by a single coefficient formula; the triple constants are $\zeta(3), 0, -\zeta(3)$ in scale order. An explicit resonant integer-scale triple has a different, fully evaluated answer. Second, a mixed derivative of a logarithmic simplex face with any number of singly labelled displaced masses reduces to a divided difference of one classical polylogarithm. Two integrations give a finite depth-three colored-polylogarithm dictionary for every three-cluster face B^r, C, D, and hence an ordinary convergent spectral formula for its Gamma analogue. Third, we prove the complete gap-ten harmonic-tail reconstruction family. An analytic argument covers every $b \geq 73728$, and an exact finite-field certificate covers all 73722 remaining indices. A general Green-function formula represents inverse coefficients by determinants of bounded size and yields ordinary convergent polygamma-product sums. Finally, classical Euler homotopy converts the inherited universal normal coefficients into nonrecursive Gamma and all-index Stieltjes antiderivative certificates. We prove local analytic convergence of both infinite-jet series and their translation identity. Proof dependencies, computational certificates, scope corrections and further research questions are recorded explicitly.

Keywords. Multiple polylogarithms; Hurwitz zeta; generalized Stieltjes constants; polygamma collisions; harmonic-tail determinants; Euler homotopy; finite parts; exact computation.

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1 Purpose, principal results and conventions

The aim is to add proved identities to the current research programme, while preserving the distinctions between an ordinary integral, a specified finite part, a formal reconstruction theorem and a numerical observation. The starting points are the manuscript and incoming directories named by the project maintainer [1, 2]. All twelve ZIP archives present in the retrieved incoming listing were downloaded and their mathematical source files inspected by topic. Ninety-two manuscript chapter files were also retrieved. This is a targeted proof and status audit of the directions used here, not a claim to have checked every line of every chapter. File hashes and the retrieval scope are supplied in `data/source_manifest.json`.

The newest source matters. In particular, the report *Polygamma Moments and Stieltjes Resonances* already completes gap eight, treats all positive monomial scales for two Stieltjes factors, and evaluates a substantial joint-polygamma range [4]. The expanded incoming version of *Universal Gamma Normal Forms* already gives the coefficient of every canonical monomial [6]. Those results are inputs here. The new work starts with the next fixed gap, additional collision factors, a third mass cluster, and the still missing nonrecursive primitive and infinite-jet convergence statements.

1.1 A map of the contributions

Direction	Result proved here	Precise remaining limit
Polygamma collisions	A finite Hermite-divided-difference correction for any number of factors and arbitrary positive shifts; Theorem 2.2.	The general merged product is retained as a specified finite part. Positive Stieltjes indices require further work.
Evaluated constants	The complete one-trigamma, centered-digamma family, Theorem 2.4; nonresonant and resonant triples.	The scale hypotheses determine which constant is selected.
Multiple mass clusters	A classical-polylogarithm divided difference for every labelled mixed derivative, and a depth-three primitive for B^r, C, D; Theorems 3.1 and 3.2.	The Gamma result is an ordinary spectral sum; no general finite arithmetic closure is proved.
Harmonic reconstruction	Every unequal even pair with gap ten has the optimal formal prefix; Theorem 4.1.	The unrestricted unequal-even-pair conjecture remains open.
Inverse coefficients	An $(m+1)$-square bordered determinant at fixed half-gap m; Theorem 5.1.	Its entries still use a finite causal recurrence.
Gamma–Stieltjes primitives	A nonrecursive finite formula for every compensator H_m and locally convergent normal and primitive series; Theorems 7.2, 7.3 and 7.4.	Sharp convergence domains, coefficient signs and arithmetic denominators remain separate questions.

The contact and Gamma-primitive arguments are analytic or algebraic proofs at every order. The gap-ten theorem has a finite exact computational component, fully exposed below. Numerical checks of polylogarithms and shifted integrals are diagnostics of the displayed formulas; they are not substitutes for the proofs.

1.2 Two representative identities

The most compact collision identity can be stated before introducing the full local correction. Let $\alpha_1 > \alpha_2 > \alpha_3 > 0$ be linearly independent over $\mathbb{Q}$, and let $r \in \{1, 2, 3\}$. Then

$$\boxed{\mathrm{CT}_{\epsilon\downarrow 0} \int_0^1 \psi'(x + \epsilon^{\alpha_r}) \prod_{j \neq r} \psi(x + \epsilon^{\alpha_j})\, dx = \begin{cases} \zeta(3), & r = 1, \\ 0, & r = 2, \\ -\zeta(3), & r = 3. \end{cases}} \tag{1.1}$$

Here the smallest shift carries the largest exponent. Every integral is ordinary for $\epsilon > 0$; the operation outside it is a specified asymptotic constant. The fully evaluated subtraction formula in Section 2 also expresses these constants as ordinary renormalized limits. No fitting of divergent coefficients is needed.

For the higher family put $\Psi = \psi + \gamma$ and $f(x) = x(\psi(1 + x) + \gamma) = \sum_{k \geq 2} (-1)^k \zeta(k) x^k$. With the analogous nonresonant ordering of N scales, the complete constant with the derivative in position r is

$$\boxed{T_{N,r} = -[x^N] \left\{ \frac{(f-1)^N}{N} + f^{N-r}(f-1)^{r-1} \right\}.} \tag{1.2}$$

Thus a single coefficient formula yields a weight-N zeta polynomial for every N and every derivative position. Its adjacent differences and its zero row sum are identities, rather than patterns inferred from the first rows.

The representative spectral identity has a different character. For the explicitly defined ordinary simplex face I_k and Gamma field G_a of Section 3, all $r, m \geq 1$ satisfy

$$\partial_{C_1} \cdots \partial_{C_m} I_{r+m}(G_a; B^r, C_1, \ldots, C_m) = (-1)^r r!\, \mathfrak{g}_{r,a,B}[C_1, \ldots, C_m], \tag{1.3}$$

where

$$\mathfrak{g}_{r,a,B}(C) = \sum_{n \geq 0} \frac{\mathrm{Li}_r((C - B)/((n + a)^2 + C))}{(n + a)^2 + C}.$$

The number of independent masses does not increase the depth of the polylogarithm in the differentiated formula. The finite three-cluster primitive reconstructs the information on the coordinate edges as well as the common mixed derivative.

1.3 Conventions used throughout

We write $\psi^{(r)}$ for the rth argument derivative of the digamma function, with $\psi^{(0)} = \psi$. The Hurwitz function is $\zeta(s, a)$, and a superscript on ζ denotes an order derivative unless an argument derivative is explicitly displayed. The Stieltjes normalization is

$$\zeta(1 + s, a) = \frac{1}{s} + \sum_{q \geq 0} \frac{(-1)^q}{q!} \gamma_q(a) s^q, \qquad \gamma_0(a) = -\psi(a). \tag{1.4}$$

These conventions agree with the standard Gamma and Hurwitz-zeta identities [8, 9]. Generalized harmonic numbers are $H_n^{(s)} = \sum_{j=1}^n j^{-s}$, with $H_0^{(s)} = 0$. The Bernoulli convention is $B_1 = -1/2$.

For a locally finite power–logarithm expansion, CT_ϵ selects the coefficient of $\epsilon^0 (\log \epsilon)^0$. The merged finite part at the lower endpoint is always taken in the displayed coordinate x: $\mathrm{CT}_{\delta\downarrow 0} \int_\delta^1$. It is not invariant under an arbitrary change of cutoff coordinate. All endpoint subtractions and all shifts are therefore kept explicit.

For divided differences, one node gives evaluation and repeated nodes give derivatives divided by the corresponding factorial. Cumulative colors for multiple polylogarithms are defined in

(3.10); this convention makes the relevant real branches and zero-color limits transparent. Mass derivatives are taken in independently labelled variables before imposing collisions.

Finally, formal rank in the harmonic reconstruction problem means rational matrix rank on declared symbols. It does not imply arithmetic independence of the numerical zeta or moment values assigned to them. The new conclusions are established relative to the inspected project sources. Classical homotopy, divided-difference and finite-part methods are credited; a bounded literature search cannot certify worldwide priority for the assembled identities.

2 Evaluated contacts for arbitrarily many polygamma factors

The latest incoming *Polygamma Moments and Stieltjes Resonances* report, its further-research question 6, asks for three independently shifted Stieltjes factors [4]. The following result answers the local-contact problem in the entire Stieltjes-index-zero sector, with any number of factors and arbitrary nonnegative argument derivative orders. Its uniform comparison permits all positive paths, including nonmonomial paths. Spectral derivatives corresponding to positive Stieltjes indices are not claimed here. The finite-part integration and divided-difference tools are classical [11, 13, 14]; the claimed contribution relative to the inspected sources is their complete singular-subset assembly and the evaluated consequences below.

2.1 The rational kernel as one Hermite divided difference

For positive nodes $a_1, \dots, a_p$, positive integers $h_1, \dots, h_p$, and $H = h_1 + \cdots + h_p$, define, for $0 \le d \le H-1$,

$$K_{\boldsymbol{h},d}(\boldsymbol{a}) = \lim_{R\to\infty} \left\{ \int_0^R \frac{x^d\,dx}{\prod_{i=1}^p (x+a_i)^{h_i}} - \mathbf{1}_{d=H-1} \log R \right\}. \tag{2.1}$$

Thus only the last numerator degree needs a finite part at infinity. For a smooth function F on the positive axis, write $[a_1^{(h_1)}, \dots, a_p^{(h_p)}]F$ for its Hermite divided difference with the node a_i repeated h_i times; its order is $H-1$. The standard confluent convention [11] gives $[a^{(H)}]F = F^{(H-1)}(a)/(H-1)!$.

Lemma 2.1 (Completely evaluated local kernel)**.** *For every indicated degree and every positive set of nodes,*

$$\boxed{K_{\boldsymbol{h},d}(\boldsymbol{a}) = (-1)^{H+d}[a_1^{(h_1)}, \dots, a_p^{(h_p)}](z^d \log z).} \tag{2.2}$$

In particular this is a finite expression in rational functions of the nodes and their logarithms, including all confluent cases.

Proof. First take H distinct nodes $b_1, \dots, b_H$ and all multiplicities equal to one. The partial fraction coefficients of $x^d/\prod_j (x+b_j)$ are

$$c_i = \frac{(-b_i)^d}{\prod_{j\ne i}(b_j - b_i)}.$$

If $d \le H-2$ their sum is zero; if $d = H-1$ their sum is one. Integrating from zero to R and subtracting the prescribed logarithm therefore gives

$$-\sum_i c_i \log b_i = (-1)^{H+d} \sum_i \frac{b_i^d \log b_i}{\prod_{j\ne i}(b_i - b_j)},$$

which is exactly the divided difference. Coalescing nodes proves the general case. For $d < H-1$ dominated convergence applies directly. For $d = H-1$, apply it to the integral on $(0,1)$ and to $x^d/\prod_j (x+b_j) - 1/x$ on $(1,\infty)$, whose tail has a locally uniform $O(x^{-2})$ bound. This also proves the confluent continuation without choosing singular individual coefficients. □

For direct implementation at distinct nodes one can instead use

$$c_{i,q} = [y^{h_i-q}]\,(y-a_i)^d \prod_{j\neq i}(y+a_j-a_i)^{-h_j}, \tag{2.3}$$

$$K_{\boldsymbol{h},d}(\boldsymbol{a}) = -\sum_i c_{i,1}\log a_i + \sum_i \sum_{q=2}^{h_i} \frac{c_{i,q}a_i^{1-q}}{q-1}. \tag{2.4}$$

Every coefficient in the first line is finite. These equations are also a transparent way to extract power–logarithm constants along specified paths.

2.2 The complete uniform local correction

Fix $N \geq 1$ and $r_1, \ldots, r_N \in \mathbb{Z}_{\geq 0}$. Put

$$h_i = r_i + 1, \qquad p_i = (-1)^{r_i+1} r_i!, \qquad g_i(x) = \psi^{(r_i)}(1+x).$$

The classical polygamma recurrence [8, Eq. 5.15.5] gives

$$\psi^{(r_i)}(x) = p_i x^{-h_i} + g_i(x). \tag{2.5}$$

For a nonempty subset $S \subseteq \{1, \ldots, N\}$ let

$$H_S = \sum_{i\in S} h_i, \qquad p_S = \prod_{i\in S} p_i, \tag{2.6}$$

$$A_{S,d}(\boldsymbol{a}) = [x^d] \prod_{j\notin S} g_j(x+a_j) = \sum_{\substack{d_j\geq 0 \ (j\notin S) \\ \sum d_j = d}} \prod_{j\notin S} \frac{\psi^{(r_j+d_j)}(1+a_j)}{d_j!}. \tag{2.7}$$

The empty product convention makes $A_{S,0} = 1$ and $A_{S,d} = 0$ for $d > 0$ when S is the full set. Crucially, the shifts in (2.7) are retained exactly.

Define the fixed-coordinate merged finite part

$$F_{\boldsymbol{r}} = \mathrm{CT}_{\delta\downarrow 0} \int_\delta^1 \prod_{i=1}^N \psi^{(r_i)}(x)\,dx \tag{2.8}$$

and the completely evaluated correction

$$\begin{aligned} Q_{\boldsymbol{r}}(\boldsymbol{a}) &= \sum_{\varnothing\neq S\subseteq\{1,\ldots,N\}} p_S \sum_{d=0}^{H_S-1} A_{S,d}(\boldsymbol{a}) K_{\boldsymbol{h}_S,d}(\boldsymbol{a}_S) \\ &= \sum_{\varnothing\neq S} \left(\prod_{i\in S} r_i!\right) \sum_{d=0}^{H_S-1} (-1)^d A_{S,d}(\boldsymbol{a})[a_i^{(h_i)} : i\in S](z^d \log z). \end{aligned} \tag{2.9}$$

Theorem 2.2 (Every polygamma collision with positive shifts)**.** *As all $a_i \downarrow 0$, with no restriction on their relative rates,*

$$\boxed{\int_0^1 \prod_{i=1}^N \psi^{(r_i)}(x+a_i)\,dx = F_{\boldsymbol{r}} + Q_{\boldsymbol{r}}(\boldsymbol{a}) + o(1).} \tag{2.10}$$

The $o(1)$ is uniform as $\max_i a_i \to 0$ through positive shifts. The same statement holds after replacing any zero-order factor ψ by $\psi + c_i$, provided the same constant is added to its regular part in (2.7) and in the merged finite part.

Proof. Expand the product by choosing its singular part on a subset S. For $S \neq \varnothing$, put

$$R_S(x;\boldsymbol{a}) = \prod_{i\in S}(x+a_i)^{-h_i}, \qquad G_S(x;\boldsymbol{a}) = \prod_{j\notin S} g_j(x+a_j).$$

Uniform Taylor expansion on a fixed compact interval gives

$$G_S(x;\boldsymbol{a}) = \sum_{d=0}^{H_S-1} A_{S,d}(\boldsymbol{a})x^d + x^{H_S}B_S(x;\boldsymbol{a}),$$

where B_S is uniformly bounded and tends to $B_S(x;0)$ for each $x \in [0,1]$. Since $R_S(x;\boldsymbol{a}) \le x^{-H_S}$ for positive shifts, the integral of $x^{H_S}R_SB_S$ tends uniformly to $\int_0^1 B_S(x;0)dx$. This estimate does not compare one shift with another.

For $d \le H_S - 2$, comparison with the tail on $(1,\infty)$ yields

$$\int_0^1 x^d R_S(x;\boldsymbol{a})dx = K_{\boldsymbol{h}_S,d}(\boldsymbol{a}_S) - \frac{1}{H_S-d-1} + o(1).$$

For $d = H_S - 1$, compare instead with $\int_1^\infty (x^{H_S-1}R_S - 1/x)dx$; this tends to zero uniformly, so the same formula holds with its rational term omitted. Both tail comparisons follow from an $O(\max a_i)x^{-2}$ bound after the appropriate leading power is removed.

The empty-subset term converges ordinarily. The limits of all remainders and all displayed rational terms sum exactly to $F_{\boldsymbol{r}}$: a merged monomial x^{d-H_S} contributes $-1/(H_S-d-1)$ for $d \le H_S-2$, while x^{-1} contributes zero to the fixed-coordinate constant on $(0,1)$. This proves (2.10). Uniform dominated convergence follows by taking the supremum over the shrinking parameter cube in the same integrable bound. Constants added to the regular parts do not change the proof. □

This theorem evaluates the entire local correction. It does not assert a new scalar evaluation of every merged product $F_{\boldsymbol{r}}$; some such merged values remain spectral or harmonic constants in the prior manuscript. The next family eliminates that baseline by an ordinary antiderivative and hence gives complete scalar identities.

2.3 Nonresonant scales and their ordering

Assume

$$a_i = \lambda_i \epsilon^{\alpha_i}, \qquad \alpha_1 > \alpha_2 > \cdots > \alpha_N > 0, \qquad \lambda_i > 0, \tag{2.11}$$

and that $\alpha_1,\dots,\alpha_N$ are linearly independent over $\mathbb{Q}$. For a nonempty subset S, write $m(S) = \max S$; its node is the largest shift within S. Let $H_0 = 0$ and $H_r = 1 + \cdots + 1/r$.

Theorem 2.3 (Evaluated nonresonant constant)**.** *The ordinary shifted integral has a power–logarithm expansion through its constant, and*

$$\boxed{\mathrm{CT}_\epsilon \int_0^1 \prod_i \psi^{(r_i)}(x+a_i)dx = F_{\boldsymbol{r}} - \sum_{\varnothing\neq S} p_S A_{S,H_S-1}(0)\big(\log\lambda_{m(S)} + H_{r_{m(S)}}\big).} \tag{2.12}$$

The formula includes the same centering modification as Theorem 2.2. Its correction is a finite polynomial in ordinary zeta values, Euler's constant and the displayed logarithms.

Proof. The rational formula (2.4) expands in convergent geometric series in ratios of smaller shifts to larger shifts. After each finite pole order is factored out, all remaining increments of the exponent are positive differences $\alpha_i - \alpha_j$ or positive regular-part exponents. Therefore

only finitely many terms can occur below any fixed power. Analytic expansion of the regular coefficients has the same property.

For $d < H_S - 1$, the rational coefficients of $K_{\boldsymbol{h}_S,d}$ are homogeneous of degree $d+1-H_S < 0$, and the logarithms are $\log \lambda_i + \alpha_i \log \epsilon$. A zero exponent would, by rational independence, require every integer exponent of every shift variable to be zero. Powers coming from regular-part shifts occur only in variables outside S, so they must all be zero. The remaining degree cannot be zero, a contradiction. Thus no noncritical degree contributes a constant.

For $d = H_S - 1$, regular-part powers outside S again cannot contribute. Put $b = a_{m(S)}$ and $\rho_i = a_i/b$. Scaling the defining integral gives

$$K_{\boldsymbol{h}_S,H_S-1}(\boldsymbol{a}_S) = -\log b + K_{\boldsymbol{h}_S,H_S-1}((\rho_i)_{i\neq m(S)}, 1).$$

Dominated convergence on $(0,1)$ and on the compensated tail gives

$$\begin{aligned}\lim_{\rho_i\to 0} K_{\boldsymbol{h}_S,H_S-1}((\rho_i),1) &= \lim_{R\to\infty}\left\{\int_0^R \frac{x^{h_{m(S)}-1}}{(1+x)^{h_{m(S)}}}dx - \log R\right\}\\ &= -H_{h_{m(S)}-1}.\end{aligned}$$

For the first dominated-convergence step use the majorant $x^{h_{m(S)}-1}/(1+x)^{h_{m(S)}}$ on $(0,1)$; the compensated tail is uniformly $O(x^{-2})$. The last evaluation follows from $t = x/(1+x)$ and $t^{h-1}/(1-t) = 1/(1-t) - \sum_{j=0}^{h-2} t^j$. Consequently the critical constant is $-\log \lambda_{m(S)} - H_{r_{m(S)}}$. Substitution into (2.9) proves the theorem. □

Rational independence is a sufficient clean hypothesis, not an assertion that every rationally dependent choice is resonant. The exact kernel formula remains valid for every positive path and permits finite coefficient extraction in every specified monomial case.

2.4 A family with complete zeta-valued answers

Let

$$\Psi(x) = \psi(x) + \gamma, \qquad h(x) = \psi(1+x) + \gamma, \qquad f(x) = xh(x) = \sum_{k\geq 2} (-1)^k \zeta(k) x^k.$$

The series is the classical digamma Taylor expansion [8, Eq. 5.7.4]. Thus $\Psi(x) = -1/x + h(x)$ and $\Psi(1) = h(0) = 0$. Take unit amplitudes in (2.11). For $1 \leq r \leq N$, define the full constant of the ordinary interval integral

$$T_{N,r} = \mathrm{CT}_\epsilon \int_0^1 \psi'(x + \epsilon^{\alpha_r}) \prod_{j\neq r} \Psi(x + \epsilon^{\alpha_j})\, dx. \tag{2.13}$$

Theorem 2.4 (Every centered one-derivative moment)**.** *For every $N \geq 2$ and $1 \leq r \leq N$,*

$$\boxed{T_{N,r} = -[x^N]\left\{\frac{(f(x)-1)^N}{N} + f(x)^{N-r}(f(x)-1)^{r-1}\right\}.} \tag{2.14}$$

This is a finite polynomial in ordinary zeta values of total weight N. It is independent of the particular nonresonant exponents and depends only on the position of the derivative in their ordering. Moreover

$$\sum_{r=1}^{N} T_{N,r} = 0, \qquad T_{N,r+1} - T_{N,r} = [x^N] f(x)^{N-r-1}(f(x)-1)^{r-1}. \tag{2.15}$$

Proof. The merged antiderivative is $\Psi(x)^N/N$, so its finite part is

$$B_N = -\frac{1}{N}[x^N](f(x)-1)^N.$$

In (2.12), only a singular subset that contains the derivative, has that derivative as its largest shift, and contains any selection of its $r-1$ smaller shifts can contribute: all other subsets have harmonic number $H_0 = 0$. A subset of size s has $H_S = s+1$, prefactor $p_S = (-1)^{s-1}$, and regular coefficient $[x^s]h(x)^{N-s}$. Its contact is therefore $(-1)^s[x^s]h(x)^{N-s}$, with multiplicity $\binom{r-1}{s-1}$. Their sum is

$$\begin{aligned} C_{N,r} &= \sum_{s=1}^{r}(-1)^s\binom{r-1}{s-1}[x^s]h(x)^{N-s} \\ &= -[x^N]f(x)^{N-r}(f(x)-1)^{r-1}. \end{aligned}$$

Adding B_N gives the theorem. The adjacent difference is immediate. For the sum use $\sum_{r=1}^{N} f^{N-r}(f-1)^{r-1} = f^N - (f-1)^N$ and $[x^N]f^N = 0$, since f starts in degree two. □

The first rows, simplified with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$, are

$$(T_{2,1}, T_{2,2}) = (0,0), \tag{2.16}$$

$$(T_{3,1}, T_{3,2}, T_{3,3}) = (\zeta(3), 0, -\zeta(3)), \tag{2.17}$$

$$(T_{4,1}, T_{4,2}, T_{4,3}, T_{4,4}) = \frac{\zeta(4)}{4}(-11,-1,5,7), \tag{2.18}$$

$$\begin{aligned} (T_{5,1},\dots,T_{5,5}) = (&\zeta(5) - 4\zeta(2)\zeta(3), \zeta(5) - 4\zeta(2)\zeta(3), \\ &\zeta(5) - 2\zeta(2)\zeta(3), 2\zeta(2)\zeta(3), 8\zeta(2)\zeta(3) - 3\zeta(5)). \end{aligned} \tag{2.19}$$

These are full asymptotic constants of the integrals, not merely their differences from the merged finite part.

2.5 The uncentered triple and an explicit resonant path

For the uncentered triple the merged value is

$$\mathrm{CT}_{\delta\downarrow 0}\int_{\delta}^{1}\psi'(x)\psi(x)^2\,dx = \zeta(3) - 2\gamma\zeta(2). \tag{2.20}$$

The three nonresonant contacts, with unit amplitudes, are respectively $2\gamma\zeta(2)$, $2\gamma\zeta(2)-\zeta(3)$, and $2\gamma\zeta(2)-2\zeta(3)$. Thus the full constants remain exactly the triple row (2.17). A concrete admissible choice is $(\alpha_1,\alpha_2,\alpha_3) = (\sqrt{3},\sqrt{2},1)$. The cyclic sum is zero, also directly forced by integrating the derivative of $\prod_i \psi(x+a_i)$ and taking its constant.

For comparison, the resonant path $(a,b,c) = (\epsilon^3,\epsilon^2,\epsilon)$ with the derivative at a has the different exact value

$$\boxed{\mathrm{CT}_{\epsilon}\int_0^1 \psi'(x+\epsilon^3)\psi(x+\epsilon^2)\psi(x+\epsilon)dx = 4 + 2\gamma - \zeta(2) + \zeta(3) + \left(\frac{9}{2}-\gamma\right)\zeta(4) + \zeta(5) - \zeta(6).} \tag{2.21}$$

Here is a finite certificate of its constant extraction. At unit amplitudes every logarithm in (2.4) is an integer multiple of $\log\epsilon$, so only its rational part can contribute to the log-free constant. Any singular subset not containing the derivative has only simple poles and hence contributes zero. The four other subsets give the following constants, where $h(x) = \psi(1+x)$:

Singular subset	Its contact constant
$\{a\}$	$-\gamma\zeta(4)+\zeta(2)^2+2\gamma\zeta(2)$
$\{a,b\}$	$\gamma-\zeta(3)+2\zeta(4)+\zeta(5)-\zeta(6)$
$\{a,c\}$	$\gamma-\zeta(2)+\zeta(3)$
$\{a,b,c\}$	4

For example the last rational part is $1/[a(b-a)(c-a)]=\epsilon^{-6}/[(1-\epsilon)(1-\epsilon^2)]$, whose constant is four. The two-factor subset with remaining regular factor $h(x+c)$ has rational part

$$-\frac{h(c)-ah'(c)+a^2h''(c)/2}{a(b-a)},$$

and the exchanged expression gives the other row. The singleton has rational part $h(b)h(c)/a-h'(b)h(c)-h(b)h'(c)$. Taking the indicated finite Taylor coefficients and adding (2.20) proves (2.21).

Ordinary renormalized limits. The symbol CT_ϵ always selects $\epsilon^0(\log\epsilon)^0$ from a proved expansion; it is not the ordinary limit of a divergent integral. All the results also give ordinary limits: subtract the finite expression $Q_{\boldsymbol{r}}(\boldsymbol{a})$ from the original integral and then add its evaluated constant. Equation (2.10) proves the resulting limit equals the stated full constant. Thus the subtraction is an explicit finite formula in polygammas, rational functions and logarithms, with no fitted coefficients or new integrals.

The exact and numerical verification records for this section are described in Section 10. The kernel methods are classical [11, 12, 13, 14]; the complete assembly and evaluated families are the continuation results relative to the inspected project sources.

3 Mixed mass derivatives and the three-cluster polylogarithm dictionary

The incoming two-cluster theorem evaluates $I_{r+s}(G_a;B^r,C^s)$, and explicitly leaves B^r,C,D with two unequal displaced masses as its next target [5]. We resolve that target by a finite dictionary of weight $r+2$ and depth at most three, uniformly in r. An accompanying theorem is stronger at the differential level: the mixed derivative in arbitrarily many singly labelled displaced masses is a divided difference of one classical polylogarithm. The spectral sum in the Gamma application remains an ordinary infinite sum; no arithmetic closure or minimal-depth claim is made.

3.1 Normalization and the inherited coefficient identity

For $k\ge 2$, write $\Sigma_k=\{t_i\ge 0:\sum_i t_i=1\}$ with measure $dt_1\cdots dt_{k-1}$. For nonempty $T\subseteq\{1,\dots,k\}$ let $t_T=\sum_{i\in T}t_i$ and $b_T=\sum_{i\in T}t_ib_i/t_T$. The perspective $t_TF(b_T)$ is filled by zero when $t_T=0$. Set

$$\mathcal{H}_k(F;t)=\sum_{\varnothing\neq T}(-1)^{k-|T|}t_TF(b_T),\tag{3.1}$$

$$I_k(F;b_1,\dots,b_k)=\int_{\Sigma_k}\mathcal{H}_k(F;t)\frac{dt}{t_1\cdots t_k}.\tag{3.2}$$

These integrals are ordinary and absolutely convergent for C^1 functions on the mass interval. The perspectives are Lipschitz and the alternating numerator vanishes on each coordinate face, so it is bounded by a constant times $\min_i t_i$. This also supplies locally integrable bounds after sufficiently small independent complex powers of the t_i are inserted. Constants and affine functions are annihilated.

Let $D_F(\boldsymbol{\alpha};\boldsymbol{b})$ be the normalized Dirichlet mean of $F(\sum t_i b_i)$, with total parameter $A=\sum_i \alpha_i$. The independent-regulator identity inherited from the two-cluster report [5] implies

$$I_k(F;\boldsymbol{b}) = [\alpha_1\cdots\alpha_k]\{AD_F(\boldsymbol{\alpha};\boldsymbol{b})\}. \tag{3.3}$$

Indeed, the normalized alternating Dirichlet integral equals $(\prod_i \alpha_i)^{-1}$ times the subset alternating sum of $A_T D_F(\boldsymbol{\alpha}_T;\boldsymbol{b}_T)$; every proper subset omits a variable and disappears under squarefree coefficient extraction. When r labels have the common mass B, their parameters aggregate into X, and hence

$$I_{r+m}(F;B^r,C_1,\dots,C_m) = r![X^r Y_1\cdots Y_m]\{AD_F(X,Y_1,\dots,Y_m)\}. \tag{3.4}$$

Here $r\ge 1$, $m\ge 1$, and $A=X+Y_1+\cdots+Y_m$. All coefficients below are taken in the holomorphic continuation of the displayed normalized mean. For the logarithms used below, its existence also follows directly by expanding around the largest positive spectral factor: the resulting Dirichlet moment series has a geometric convergence margin and denominators $(1+A)_{n-1}$.

3.2 Arbitrarily many mixed derivatives reduce to one classical polylogarithm

For an analytic function f, its divided difference [11] is denoted by $f[C_1,\dots,C_m]$; thus one node means $f(C_1)$ and two nodes mean $(f(C_1)-f(C_2))/(C_1-C_2)$. Repeated nodes are assigned their unique confluent values, so $f[C,\dots,C]=f^{(m-1)}(C)/(m-1)!$.

Theorem 3.1 (Classical-polylogarithm mixed derivatives)**.** *Let $\sigma>0$, $B,C_1,\dots,C_m\ge 0$, $r,m\ge 1$, and put*

$$g_{r,\sigma,B}(C) = \frac{1}{\sigma+C}\,\mathrm{Li}_r\!\left(\frac{C-B}{\sigma+C}\right). \tag{3.5}$$

Then

$$\boxed{\partial_{C_1}\cdots\partial_{C_m} I_{r+m}(U\mapsto -\log(\sigma+U);B^r,C_1,\dots,C_m) = (-1)^r r!\, g_{r,\sigma,B}[C_1,\dots,C_m].} \tag{3.6}$$

The derivative is taken in each independently labelled displaced mass before any nodes are made equal. In particular, the bound of one on classical polylogarithm depth is independent of both r and m.

Proof. Write $S_B=\sigma+B$ and $S_j=\sigma+C_j$. Differentiate the Dirichlet mean in (3.4). Since

$$\partial_{C_1}\cdots\partial_{C_m}[-\log(S_B t_0+\sum_j S_j t_j)] = (-1)^m(m-1)!\frac{t_1\cdots t_m}{(S_B t_0+\sum_j S_j t_j)^m},$$

the coefficient of $Y_1\cdots Y_m$, first at $\Re X>0$, is

$$(-1)^m(m-1)!X\int_{\Sigma_{m+1}}\frac{t_0^{X-1}}{(S_B t_0+\sum_j S_j t_j)^m}\,dt.$$

This follows either by shifting each Y_j by one in the Dirichlet density or directly from $1/\Gamma(Y_j)=Y_j+O(Y_j^2)$. At a fixed $t=t_0$, the ordinary simplex resolvent integral gives

$$\int_{\Sigma_m}\frac{du}{(S_B t+(1-t)\sum_j S_j u_j)^m} = \frac{1}{(m-1)!\prod_{j=1}^m (S_B t+(1-t)S_j)}.$$

For completeness, this is the positive-parameter Feynman identity: write the product of m reciprocal positive numbers as a Laplace integral over the positive orthant and separate its radial variable. Therefore the required coefficient of the differentiated mean is

$$(-1)^m X\int_0^1 \frac{t^{X-1}(1-t)^{m-1}}{\prod_j (S_B t+(1-t)S_j)}\,dt. \tag{3.7}$$

At fixed t, the divided difference of $C \mapsto (S_B t + (1-t)(\sigma + C))^{-1}$ is $(-1)^{m-1}(1-t)^{m-1}$ divided by the product in (3.7). Thus that expression is the negative divided difference of

$$R_X(C) = X \int_0^1 \frac{t^{X-1}}{S_B t + (1-t)(\sigma + C)}\, dt.$$

Writing $z = (C-B)/(\sigma + C)$ yields, initially for $|z| < 1$,

$$R_X(C) = \frac{1}{\sigma + C}\left(1 + \sum_{n=1}^{\infty} \frac{X z^n}{X+n}\right), \qquad [X^r]R_X(C) = \frac{(-1)^{r-1}}{\sigma + C}\,\mathrm{Li}_r(z). \tag{3.8}$$

The series is normally convergent near $X = 0$ on each compact z-disk. Its integral representation continues it through every real $z < 1$ without crossing a singularity; equivalently, subtract the value at $t = 0$ before continuing the X integral. Now take $r![X^r]$ and use (3.4). All mass derivatives and confluent limits are justified by local holomorphy away from nonpositive spectral factors. This proves the theorem. □

For $m = 1$, integration from $C = B$ recovers the inherited one-displaced formula

$$I_{r+1}(-\log(\sigma + U); B^r, C) = (-1)^r r!\,\mathrm{Li}_{1,r}\left(\frac{C-B}{\sigma + C}\right), \tag{3.9}$$

because $d\,\mathrm{Li}_{1,r}(z)/dz = \mathrm{Li}_r(z)/(1-z)$ and the face vanishes when all masses coincide. This recovery checks the factorial and sign conventions.

3.3 A uniform finite dictionary for the three-cluster primitive

It is useful to specify colored multiple polylogarithms by cumulative colors, which avoids apparent quotients by zero. For positive indices $s_1, \dots, s_d$, define

$$\mathcal{C}_{s_1,\dots,s_d}(c_1, \dots, c_d) = \sum_{n_1 > \cdots > n_d \ge 1} \frac{c_1^{n_1 - n_2} c_2^{n_2 - n_3} \cdots c_{d-1}^{n_{d-1} - n_d} c_d^{n_d}}{n_1^{s_1} \cdots n_d^{s_d}}. \tag{3.10}$$

For $|c_j| < 1$ this series converges absolutely. In the conventional colored notation it is $\mathrm{Li}_{s_1,\dots,s_d}(c_1, c_2/c_1, \dots, c_d/c_{d-1})$, interpreted continuously when a color is zero. The one-variable convention used by the incoming reports is $\mathrm{Li}_{s_1,\dots,s_d}(z) = \mathcal{C}_{s_1,\dots,s_d}(z, \dots, z)$. For real cumulative colors below one, the continuation is the iterated-integral branch continued from zero, with letters $0, 1/c_1, \dots, 1/c_d$ and the path $[0,1]$. No letter crosses that path on the stated real domain.

Put $H(z) = \mathrm{Li}_1(z) = -\log(1-z)$ and

$$\mathcal{S}_r(z,w) = \mathcal{C}_{1,1,r}(z,w,w) + \mathcal{C}_{1,r,1}(z,z,w) + \mathcal{C}_{1,r+1}(z,w), \tag{3.11}$$

$$\mathcal{K}_r(z,w) = H(w)\,\mathrm{Li}_{1,r}(z) + H(z)\,\mathrm{Li}_{1,r}(w) - \mathcal{S}_r(z,w) - \mathcal{S}_r(w,z), \tag{3.12}$$

$$\mathcal{P}_r(z,w) = (r+1)\{\mathrm{Li}_{1,r+1}(z) + \mathrm{Li}_{1,r+1}(w)\} + \mathcal{K}_r(z,w). \tag{3.13}$$

Every term has weight $r+2$; the products in (3.12) can be shuffled as iterated integrals into depth at most three, retaining the same safe letters for real negative colors. Thus the formula gives a finite dictionary with an upper bound of three on depth for every $r \ge 1$.

Theorem 3.2 (Three-cluster logarithmic face)**.** *For $\sigma > 0$, $B, C, D \ge 0$, $r \ge 1$, let $z = (C-B)/(\sigma + C)$ and $w = (D-B)/(\sigma + D)$. Then*

$$\boxed{I_{r+2}(-\log(\sigma + U); B^r, C, D) = (-1)^{r+1} r!\,\mathcal{P}_r(z,w).} \tag{3.14}$$

In particular, if $B \le C, D$, all cumulative colors in the finite dictionary lie in $[0,1)$, regardless of the sizes of the mass displacements. The result therefore requires no small-displacement hypothesis.

Proof. The key harmonic identity, valid for $|z|, |w| < 1$, is

$$\mathcal{K}_r(z,w) = \sum_{m,n\geq 1} \frac{z^m w^n}{mn}\{H^{(r)}_{m-1} + H^{(r)}_{n-1} - H^{(r)}_{m+n-1}\}. \tag{3.15}$$

To check its finite reduction, the first two harmonic terms give the two products in (3.12). For the last term use $1/(mn) = (1/m + 1/n)/(m+n)$ and put $N = m+n$. The harmonic index $j < N$ can lie below m, above m, or equal m; these three cases give respectively $\mathcal{C}_{1,1,r}(z,w,w)$, $\mathcal{C}_{1,r,1}(z,z,w)$, and $\mathcal{C}_{1,r+1}(z,w)$. Exchanging z, w supplies the other half. All rearrangements are absolute.

Write $f_r(z) = (1-z)\operatorname{Li}_r(z)$. Differentiating (3.15) gives

$$\partial_z \partial_w \mathcal{K}_r(z,w) = -\frac{f_r[z,w]}{(1-z)(1-w)}. \tag{3.16}$$

Indeed, $\sum_{n\geq 1} H^{(r)}_{n-1} z^{n-1} = \operatorname{Li}_r(z)/(1-z)$, while the diagonal sum involving $H^{(r)}_{m+n-1}$ equals

$$\frac{\operatorname{Li}_r(z)/(1-z) - \operatorname{Li}_r(w)/(1-w)}{z-w}.$$

Combining these expressions proves (3.16), including its removable diagonal.

Now $z' = (1-z)/(\sigma + C)$, $w' = (1-w)/(\sigma + D)$,

$$z - w = \frac{(\sigma+B)(C-D)}{(\sigma+C)(\sigma+D)}, \qquad f_r(z) = (\sigma+B) g_{r,\sigma,B}(C).$$

Consequently the mixed C, D derivative of the right side of (3.14) agrees exactly with Theorem 3.1 for $m = 2$. On $C = B$ or $D = B$, $\mathcal{K}_r$ vanishes and the remaining term is precisely the inherited one-displaced formula with $r+1$ base labels. Equality follows from two integrations of the common mixed derivative, starting on those two coordinate edges. The proof initially applies in a neighborhood of $C = D = B$ and throughout $B \leq C, D$ by the absolutely convergent color series. Analytic continuation along the positive mass domain gives every stated $B, C, D \geq 0$. □

The formula is symmetric under $C \leftrightarrow D$. Its two principal collision checks are

$$\mathcal{P}_r(z,0) = (r+1)\operatorname{Li}_{1,r+1}(z), \tag{3.17}$$

$$\mathcal{P}_r(z,z) = 2r\operatorname{Li}_{1,r+1}(z) + 2\sum_{j=1}^{r} \operatorname{Li}_{1,j,r+1-j}(z). \tag{3.18}$$

The second identity recovers twice the predecessor's $P_{r,2}$. It also follows directly from the shuffle identity

$$H(z)\operatorname{Li}_{1,r}(z) = \operatorname{Li}_{1,1,r}(z) + \operatorname{Li}_{1,r,1}(z) + \sum_{j=1}^{r} \operatorname{Li}_{1,j,r+1-j}(z).$$

Thus the three-cluster formula has the correct labelled multiplicity even when the two displaced masses become equal.

3.4 Gamma faces, confluent derivatives, and polygamma values

For $a > 0$, $U \geq 0$, put

$$G_a(U) = \log\Gamma(a + i\sqrt{U}) + \log\Gamma(a - i\sqrt{U}) - \log(2\pi).$$

The analytic logarithms are continued from positive real Gamma arguments. The ordinary Gamma product [8, §5.8] gives

$$G_a(U) = 2\log\Gamma(a) - \log(2\pi) - \sum_{n=0}^{\infty} \log\left(1 + \frac{U}{(n+a)^2}\right).$$

Its second mass derivative is normally convergent on compact nonnegative mass sets, so every ordinary face may be applied termwise after subtracting an affine function from each term.

Corollary 3.3 (The three-cluster Gamma dictionary)**.** *For $a > 0$, $B, C, D \geq 0$ and $r \geq 1$, set*

$$z_n = \frac{C - B}{(n+a)^2 + C}, \qquad w_n = \frac{D - B}{(n+a)^2 + D}.$$

Then

$$\boxed{I_{r+2}(G_a; B^r, C, D) = (-1)^{r+1} r! \sum_{n=0}^{\infty} \mathcal{P}_r(z_n, w_n).} \tag{3.19}$$

The sum is ordinary and absolutely convergent, locally normally in the stated parameter domain. Each spectral term has a finite colored multiple-polylogarithm representation of depth at most three.

Proof. Apply Theorem 3.2 termwise to the Gamma product. For a direct tail bound, (3.15) gives

$$\mathcal{P}_r(z, w) = \frac{r+1}{2}(z^2 + w^2) - zw + O((|z| + |w|)^3).$$

Both colors are $O(n^{-2})$, so the spectral summands are $O(n^{-4})$. Compact local holomorphic extensions give the same normal convergence for every fixed parameter derivative. □

Define the normally convergent classical-polylogarithm sum

$$\mathfrak{g}_{r,a,B}(C) = \sum_{n=0}^{\infty} \frac{\mathrm{Li}_r((C - B)/((n+a)^2 + C))}{(n+a)^2 + C}. \tag{3.20}$$

Theorem 3.1 gives, for every $m \geq 1$,

$$\boxed{\partial_{C_1} \cdots \partial_{C_m} I_{r+m}(G_a; B^r, C_1, \ldots, C_m) = (-1)^r r! \mathfrak{g}_{r,a,B}[C_1, \ldots, C_m].} \tag{3.21}$$

For example, the first case beyond the two-cluster primitive is

$$\partial_C \partial_D I_4(G_a; B, B, C, D) = \frac{2}{C - D} \sum_{n=0}^{\infty} \left\{ \frac{\mathrm{Li}_2(z_n)}{(n+a)^2 + C} - \frac{\mathrm{Li}_2(w_n)}{(n+a)^2 + D} \right\}. \tag{3.22}$$

At $C = D$ it is the corresponding derivative of $\mathfrak{g}_{2,a,B}$; no singular mass denominator remains.

More generally, making m independent displaced masses equal after differentiation gives

$$\partial_{C_1} \cdots \partial_{C_m} I_{r+m}(G_a; B^r, C_1, \ldots, C_m)|_{C_j = C} = \frac{(-1)^r r!}{(m-1)!} \partial_C^{m-1} \mathfrak{g}_{r,a,B}(C). \tag{3.23}$$

Every finite derivative here involves only classical polylogarithms Li_{r-j} and rational mass factors. Negative polylogarithm orders are rational functions. The distinction between these labelled mixed derivatives and total derivatives along a diagonal is essential.

At the complete mass collision $C = B$, the formula has a finite harmonic-polygamma specialization. Let $H^\star_n(\{1\}^r) = \sum_{n\ge j_1\ge\cdots\ge j_r\ge 1}(j_1\cdots j_r)^{-1}$. For $m \ge 2$,

$$\partial_{C_1}\cdots\partial_{C_m} I_{r+m}(G_a; B^r, C_1, \dots, C_m)|_{C_j=B} = (-1)^{r+m} r! H^\star_{m-1}(\{1\}^r) \sum_{n=0}^{\infty} ((n+a)^2 + B)^{-m}. \tag{3.24}$$

Indeed the coefficient of u^N in $(1+u)^{-1}\operatorname{Li}_r(u/(1+u))$ is $(-1)^{N+1}H^\star_N(\{1\}^r)$; this follows from the elementary binomial identity $\sum_{j=1}^{N}(-1)^{j-1}\binom{N}{j}/j^r = H^\star_N(\{1\}^r)$. At $B = 0$ the final sum is $\zeta(2m, a)$; at $B > 0$ it is a finite derivative of the paired digamma resolvent

$$\sum_{n=0}^{\infty}((n+a)^2 + B)^{-1} = \frac{\psi(a + i\sqrt{B}) - \psi(a - i\sqrt{B})}{2i\sqrt{B}}.$$

This is a consistency check with the predecessor's all-degree collision formula, not a separate novelty claim for those coefficients.

3.5 An independent ordinary integral for verification

The finite dictionary also admits a single scalar logarithmic integral that is convenient for numerical verification. Put $q = (B - C)/(\sigma + B)$, $p = (B - D)/(\sigma + B)$, $L(t) = \log(1 - t)$, $a(t) = -\log(1 - qt)$, and $b(t) = -\log(1 - pt)$. Then

$$I_{r+2}(-\log(\sigma + U); B^r, C, D) = \int_0^1 \Big[L^r\{a'(L + b) + b'(L + a)\} + (r + 1)\frac{L^r(a + b)}{t} + r\frac{L^{r-1}ab}{t}\Big]dt. \tag{3.25}$$

All apparent singularities at zero are removable or integrable; at one only powers of $\log(1 - t)$ occur. One derivation uses independent Gamma variables and Frullani's formula to write

$$AD_{-\log}(X, Y, Z) = \mathcal{L}(X, Y, Z) + \int_0^1 (1 - t)^{X+Y+Z}\frac{d}{dt}\frac{(1 - qt)^{-Y}(1 - pt)^{-Z} - 1}{t}\,dt.$$

Here $\mathcal{L}$ is linear in X, Y, Z. Extract $r![X^rYZ]$ and integrate the double-pole term using the primitive $(1 - t)/t$, which vanishes at $t = 1$. This produces (3.25) without endpoint finite parts.

For an additional check wholly in the local mass series, let $h_j(N) = H^\star_{N-1}(\{1\}^j)$. When $|q|, |p| < 1$, the independent Dirichlet moment expansion gives

$$I_{r+2}(-\log(\sigma + U); B^r, C, D) = (-1)^r r!\Big[\sum_{m,n\ge 1}\frac{q^m p^n}{mn}h_r(m + n) - (r + 1)\sum_{m\ge 1}\frac{q^m + p^m}{m}h_{r+1}(m)\Big]. \tag{3.26}$$

The two sums arise respectively from terms using both displaced Dirichlet parameters and from terms using just one. This provides an independently organized harmonic computation of the same face.

The supplied verifier records 231 assertions: 192 exact rational binomial–harmonic checks, fourteen comparisons from seven scalar parameter cases, 24 independently computed mixed-derivative checks, and one Gamma spectral check. The scalar cases include $r = 1, 2, 3, 6$, a displaced mass below the base, an equal-outlier collision, and a zero displacement. At $\sigma = 1$, $B = 0$, $C = 1/5$, $D = 2/5$, $r = 2$, for example, the common scalar value is

$$-0.30790320335103038854259002091657853760540570131340 3\ldots.$$

The finite colored dictionary, the ordinary scalar integral, and the local Dirichlet series agree at 75-digit working precision with scaled residuals below $1.5 \cdot 10^{-75}$. The independent mixed-derivative checks use the predecessor's monomial-face formula and include both complete and partial confluence; their largest scaled residual is below $6 \cdot 10^{-72}$. The Gamma example is

$$I_4(G_1; 0, 0, 1/5, 2/5) = -0.34118105526111796685557807452336322083216220144833 86\ldots. \tag{3.27}$$

A finite colored-MPL spectral head plus an exact-form Hurwitz tail agrees with the independent complete Hurwitz collision series to a scaled residual below $1.4 \cdot 10^{-76}$. These are floating-point diagnostics, not interval certificates or substitutes for the analytic proofs.

4 The complete gap-ten reconstruction theorem

The latest incoming report *Polygamma Moments and Stieltjes Resonances* proves the complete gap-eight family and gives a computable eventual threshold at every fixed gap [4]. Thus the next unresolved fixed-gap family in the inspected material is the gap of ten. We settle that entire family. The analytic improvement is to keep different error budgets for odd and even Gamma modes; only two of the five modes require the larger error. The finite proof uses exact fast polynomial arithmetic, so a substantially longer interval can be checked without quadratic propagation of the fundamental array.

We also give a bounded-size formula for every inverse coefficient at an arbitrary fixed gap. At gap ten it uses determinants of order six, independently of the two tail orders. This supplies a constructive answer to one version of the earlier compact-inverse-row question [7]. It retains a finite causal recurrence and does not assert a short product evaluation of those coefficients. These advances are relative to the inspected project sources; no claim of historical priority is made.

4.1 The original harmonic-tail coordinates

Let $b \geq 6$, $p = 2b - 10$, $q = 2b$, and

$$A_v(n) = \zeta(v, n+1) = \zeta(v) - H_n^{(v)}, \qquad x_n = n + \tfrac{1}{2}.$$

The series

$$D_{p,q}(s) = \sum_{n \geq 0} A_p(n) A_q(n) x_n^{-s}$$

converges absolutely for $\operatorname{Re} s > 3 - p - q$ and has a meromorphic continuation. Write $D_r = D_{p,q}(-2r)$. All these values are regular: the centered Hurwitz expansion

$$A_v(n) \sim \sum_{h \geq 0} c_{v,h} x_n^{1-v-2h}, \qquad c_{v,0} = \frac{1}{v-1}, \quad c_{v,h} = \frac{B_{2h}(1/2)}{(2h)!}(v)_{2h-1} \quad (h \geq 1) \tag{4.1}$$

allows subtraction of successive terms, and the resulting possible poles of $D_{p,q}$ are the odd integers $3 - p - q - 2h$. The Bernoulli convention is $B_1 = -1/2$, and $(v)_k$ is a rising factorial.

The inherited block transform is

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r, \tag{4.2}$$

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j. \tag{4.3}$$

For $j \geq b$ it has the finite late-coordinate form

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \qquad Z_k = \zeta(2k), \qquad Z_0 = -\tfrac{1}{2}. \tag{4.4}$$

For $d = 2j+1$ and $e = d-p-q+2k+1$, put $a_{jk} = 0$ if $e < 0$; otherwise the exact coefficient is

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}. \tag{4.5}$$

Each summand is omitted unless its lower binomial index is between 2 and its upper index. In particular, a zero lower index is omitted. This is the exact convention in the original source, and it will be retained throughout the proof.

For context, the finite summation underlying these formulas follows from

$$A_p(n-1)A_q(n-1) - A_p(n)A_q(n) = n^{-p}A_q(n) + n^{-q}A_p(n) + n^{-p-q}.$$

Multiply by n^{2j+1} and sum by parts. The constant term at $N \to \infty$ of

$$N^{2j+1}A_p(N)A_q(N) + \sum_{n=1}^{N} n^{2j+1-q}A_p(n) + \sum_{n=1}^{N} n^{2j+1-p}A_q(n) + \sum_{n=1}^{N} n^{2j+1-p-q}$$

is C_j. Expanding the centered difference $(x+1/2)^{2j+1} - (x-1/2)^{2j+1}$ gives (4.2); the inverse generating factor is $z/(2\sinh(z/2))$, giving (4.3). Finite Faulhaber summation gives (4.5). These coordinate identities and their constant-term interpretation are inherited from the ordered-boundary development [3].

Define the original b-square matrix

$$M_{b,5} = (a_{b+r,k})_{0 \leq r,k < b}.$$

On the specified formal labels

$$C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}, \tag{4.6}$$

the first $2b$ block rows have coefficient matrix $\operatorname{diag}(I_b, M_{b,5})$. The diagonal of (4.3) is $1/(2r+1)$, so the corresponding coefficient determinant of $D_0, \ldots, D_{2b-1}$ is

$$\frac{\det M_{b,5}}{(4b-1)!!}. \tag{4.7}$$

Formal rank below always means rational matrix rank on (4.6). It does not assert independence after these labels are evaluated as numerical constants.

Theorem 4.1 (Every gap-ten pair has the optimal prefix)**.** *For every integer $b \geq 6$, the matrix $M_{b,5}$ is nonsingular. Consequently, for every even $p \geq 2$ and $q = p+10$, the first q centered moments*

$$D_{p,q}(0), D_{p,q}(-2), \ldots, D_{p,q}(-2(q-1))$$

have formal rank q and generate all later centered even moments over $\mathbb{Q}$. The number q is optimal for the q specified formal coordinates. Their coefficient determinant is (4.7)*.*

4.2 A rational recurrence of fixed dimension

Put

$$\begin{aligned}
&\beta_n = \frac{2|B_{2n}|}{(2n)!} \quad (n \ge 1),\\
&a_u = 0 \quad (1 \le u \le 5), \qquad a_u = (2u)^{\underline{10}} \quad (u > 5),\\
&b_s = (2s-1)^{\underline{10}}, \qquad d_s = 2(a_s + b_s) \quad (s > 5).
\end{aligned}$$

The underline denotes a falling factorial; the symbols b_s here are recurrence factors, not Bernoulli numbers. Every d_s is positive. Define the rational array

$$\begin{aligned}
R_{u,j} &= \delta_{uj} && (1 \le u, j \le 5),\\
R_{s,j} &= \frac{1}{d_s}\sum_{u=1}^{s-1} \beta_{s-u}(b_s + a_u) R_{u,j} && (s > 5),
\end{aligned} \tag{4.8}$$

and let

$$F_{b,5} = (R_{b+i,j})_{1 \le i,j \le 5}, \qquad f_{b,5} = \det F_{b,5}.$$

The inherited complementary-minor identity is

$$\det M_{b,5} = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{s=6}^{b+5} d_s\right) f_{b,5}. \tag{4.9}$$

Its normalization is important. For an even series $R(z) = \sum_{n \ge 1} r_n z^{2n}$, the coefficient of z^{2s-11} in

$$\mathcal{T}_5 R = (R^{(10)} - R^{(10)}(0)) \cot(z/2) + (R \cot(z/2))^{(10)} \tag{4.10}$$

is

$$d_s r_s - \sum_{u<s} \beta_{s-u}(b_s + a_u) r_u.$$

The assignment $a_5 = 0$ records the subtracted derivative, equivalently the omitted lower binomial index zero in (4.5). Truncate the five fundamental null series after degree $2b$ and apply the complementary-minor identity to this coefficient operator. Reversing the original b columns and restoring their factorial scales gives exactly (4.9); a full normalization proof is in [3, Gamma-basis section].

4.3 The inherited Gamma basis, with the normalization made explicit

Let

$$W(x) = \left|\Gamma\left(\tfrac{1}{4} + \tfrac{ix}{2}\right)\right|^{20}, \qquad P_j(x) = \sinh(\pi(5-j)x) \ (1 \le j < 5), \qquad P_5(x) = \tanh(\pi x).$$

Write

$$t_{n,j} = \frac{4}{(2n)!} \operatorname{Im} \int_0^\infty (x + i/2)^{2n} P_j(x) W(x)\, dx. \tag{4.11}$$

These are absolutely convergent ordinary integrals. The Gamma-basis theorem [3] states that

$$G_j(z) = 4 \sin(z/2) \int_0^\infty \sinh(xz) P_j(x) W(x)\, dx$$

is a null germ of (4.10), and that

$$(t_{b+i,j})_{i,j=1}^5 = F_{b,5} T_5, \qquad T_5 = (t_{n,j})_{n,j=1}^5, \qquad \det T_5 > 0. \tag{4.12}$$

Here is the analytic mechanism, to specify precisely what is inherited. For an allowed multiplier P, set

$$Q_P(t) = 2\sinh(t/2)\int_{\mathbb{R}} e^{ixt}P(x)W(x)\,dx.$$

The Fourier contours in the two terms obtained by differentiating Q_P ten times can be shifted by $\pm i$. Gamma recurrence gives

$$(iy-\tfrac12)^{10}W(y+i) = (iy+\tfrac12)^{10}W(y-i) = 2^{10}\left[\Gamma(\tfrac34+iy/2)\Gamma(\tfrac34-iy/2)\right]^{10}.$$

The nearest poles of the common Gamma factor are at $\pm 3i/2$, outside the two shifted contours. Also $P(y+i)=P(y-i)$. The sinh modes therefore have no contact term. For tanh the two simple poles at $y=\pm i/2$, each with residue $1/\pi$, give $Q_P^{(10)}(0) = i2^{12}\pi^5$; the other four top derivatives vanish. The resulting identity is the hyperbolic version of (4.10). Rotation $G_j(z) = iQ_{P_j}(iz)$ gives the stated null equation. Stirling decay justifies all contour shifts and differentiations: on the vertical boundaries the decay is at least $e^{-\pi|\operatorname{Re} y|}$ times a fixed polynomial.

For the sign, $\operatorname{Im}(x+i/2)^{2n}$ has leading term nx^{2n-1}. A triangular row operation and continuous Cauchy–Binet express $\det T_5$ as a positive constant times

$$\int_{(0,\infty)^5} \Delta(x_1^2,\dots,x_5^2)\Delta(\cosh\pi x_1,\dots,\cosh\pi x_5)\prod_{r=1}^{5} x_r W(x_r)\tanh(\pi x_r)\,d\mathbf{x}.$$

Indeed $\sinh(ku)/\tanh u = \cosh u\, U_{k-1}(\cosh u)$, so the multiplier alternant has column degrees $4,3,2,1,0$; their reversal has positive sign. Both Vandermondes change sign together under any interchange, and their product is strictly positive off the diagonals. This proves $\det T_5 > 0$. The coefficient identity in (4.12) then follows from uniqueness in (4.8).

4.4 An effective phase estimate at half-gap five

Put

$$\lambda_j = j\pi, \qquad A = 2^5(2\pi)^{10}, \qquad a_j^* = A/2\ (j<5), \qquad a_5^* = A.$$

Lemma 4.2 (Specialization of the effective Gamma estimate)**.** *For every integer $N \ge 147456$ and $1 \le j \le 5$,*

$$\left|\frac{1}{a_j^*}\int_0^\infty (x+i/2)^N P_j(x)W(x)\,dx - e^{i\lambda_j/2}\left(\frac{\Gamma(N-4)}{\lambda_j^{N-4}} + \frac{5i}{2}\frac{\Gamma(N-5)}{\lambda_j^{N-5}}\right)\right| < 33\frac{\Gamma(N-6)}{\lambda_j^{N-6}}. \quad (4.13)$$

Proof. This is the $m=5$ case of the effective phase lemma of [4]; a specialized proof follows. For $x \ge 10$, complex Stirling with its remainder [8, §5.11(ii)] gives

$$\frac{P_j(x)W(x)}{a_j^*} = x^{-5}e^{-\lambda_j x}(1+\delta_j(x)), \qquad |\delta_j(x)| \le 10x^{-2}. \quad (4.14)$$

To check the constants, for $z = 1/4 + ix/2$ the remainder after $1/(12z)$ has magnitude at most $1/(90|z|^3)$. With $t = 1/(2x)$, the logarithm of $W/(Ax^{-5}e^{-5\pi x})$ is

$$-\frac52\log(1+t^2) + 5\left(\frac{\arctan t}{t} - 1\right) + \frac{20}{3}\frac{t^2}{1+t^2} + 20\operatorname{Re}\rho(z).$$

Its absolute value is at most $65/(24x^2) + 16/(9x^3) < 15/(4x^2)$. Using $|e^u - 1| \le 2|u|$ in this range, and the bound $2e^{-2\pi x} \le 1/(9x^2)$ for the normalized multiplier, proves (4.14). On $0 \le x \le 10$ one may use the generous bound $P_j(x)W(x)/a_j^* < 2^{460}$, obtained from $\Gamma(1/4) < 5$, $\pi < 4$, and $e < 4$.

Write $S_2 = \Gamma(N-6)/\lambda_j^{N-6}$. The density error on $x \geq \sqrt{N}$ is at most $20S_2$ because

$$|x + i/2|^N \leq x^N \exp(N/(8x^2)) < 2x^N.$$

On $10 \leq x \leq \sqrt{N}$ its bound is $\sqrt{N}(N+1)^{N/2}$. For the model integral on $[10, \infty)$, put $y = x + i/2$ and shift the ray down to the positive real axis. No pole is crossed. The algebraic factor becomes $(1 - i/(2y))^{-5}$, and Taylor's theorem, for $y \geq 10$, gives

$$\left|(1 - i/(2y))^{-5} - 1 - \frac{5i}{2y}\right| \leq \frac{45}{4y^2}.$$

For example the positive Taylor remainder is at most $15u^2(1-u)^{-7} < 45u^2$ for $u = 1/(2y) \leq 1/20$. Thus this part of the model error is at most $(45/4)S_2$.

The original compact integral, the short vertical segment, and the two retained model integrals from 0 to 10 are together bounded by $2^{720}15^N$. To absorb the remaining terms, set $k = N - 7$. Since $\lambda_j < 20$ and $k! \geq (k/3)^k$,

$$S_2 > \frac{1}{20}\left(\frac{k}{60}\right)^k.$$

For $N \geq 147456$, $k \geq 3N/4$ and $\sqrt{N+1} \leq (5/4)\sqrt{N}$, so

$$\frac{\sqrt{N}(N+1)^{N/2}}{S_2} \leq N^9(100/\sqrt{N})^N < N^9 2^{-N} \leq \tfrac{1}{4},$$
$$\frac{2^{720}15^N}{S_2} \leq 2^{720}N^8(1200/N)^N < 2^{720}N^8 4^{-N} < \tfrac{1}{4}.$$

Here $N^9 \leq 2^{N-2}$ holds at 147456 and persists by induction. The total error is less than $(20 + 45/4 + 1/2)S_2 < 33S_2$, proving (4.13). □

4.5 Different determinant budgets for odd and even modes

Proposition 4.3 (A positive tail beginning at an explicit index)**.** *For every integer $b \geq 73728$, one has $f_{b,5} > 0$.*

Proof. Apply Lemma 4.2 at $N_i = 2b + 2i$, $1 \leq i \leq 5$. An odd column has leading imaginary phase ± 1 and relative error bounded by

$$\frac{33\lambda_j^2}{(N_i - 5)(N_i - 6)} < \frac{8448}{(N_i - 6)^2},$$

since $5\pi < 16$. An even column has zero leading imaginary phase. Its first nonzero term is the second term in (4.13); its relative error is at most

$$\frac{66\lambda_j}{5(N_i - 6)} < \frac{858}{5(N_i - 6)},$$

because the largest even mode is $4\pi < 13$.

Extract the positive row factors

$$\frac{4\Gamma(N_i - 4)}{\Gamma(N_i + 1)}\pi^{-2i}$$

and positive column factors $a_j^* \lambda_j^{-2b+4}$. In an even column extract additionally $5\lambda_j/(2(2b-3))$. The five extracted signs are $(+, -, -, +, +)$, with positive product. The remaining matrix is $E_{ij} = j^{-2i}(1 + \eta_{ij})$. For its three odd columns,

$$|\eta_{ij}| < \frac{8448}{(2b-4)^2} < 2^{-20}.$$

For the two even columns the additional row factor is $\rho_i = (2b-3)/(2b+2i-5)$; hence

$$|\eta_{ij}| < \frac{8}{2b-3} + \frac{858}{5(2b-4)} < \frac{898}{5(2b-4)} < 2^{-9}.$$

Both final inequalities hold already at $b = 73728$ and improve monotonically thereafter.

For $V = (j^{-2i})_{i,j=1}^{5}$, exact rational arithmetic gives

$$\det V = \frac{49}{2764800000000}, \qquad \operatorname{perm} V = \frac{106852831033}{25798901760000000000}, \qquad \frac{\det V}{\operatorname{perm} V} = \frac{457228800}{106852831033}. \tag{4.15}$$

Every permutation product uses exactly two even columns and three odd columns. Therefore

$$\begin{aligned} |\det E - \det V| &< \operatorname{perm} V \left\{ (1+2^{-9})^2 (1+2^{-20})^3 - 1 \right\} \\ &< \frac{\operatorname{perm} V}{240} < \det V. \end{aligned}$$

The two final inequalities are exact rational inequalities. In particular $\det E > 0$. The extracted row and column factors, the product of the extracted signs, and $\det T_5$ are positive. Equation (4.12) proves $f_{b,5} > 0$. □

The distinction between the two error budgets is substantive: using the worst entrywise error five times gives a weaker threshold. The general estimate in the predecessor was $B(5) = 635656$; the preceding argument reduces the required finite interval to $b < 73728$, while using the same inherited phase remainder.

4.6 An exact certificate for every remaining integer

The cotangent coefficients satisfy

$$\beta_1 = \frac{1}{6}, \qquad 2(2n+1)\beta_n = \sum_{k=1}^{n-1} \beta_k \beta_{n-k} \quad (n \geq 2). \tag{4.16}$$

This follows from the Riccati equation for $(z/2)\cot(z/2)$. We use the prime

$$\ell = 998244353 = 119 \cdot 2^{23} + 1.$$

The supplied verifier proves primality by trial division. It also verifies that 3 is a primitive root by checking the three prime divisors $2, 7, 17$ of $\ell - 1$. Every polynomial transform used in the certificate has length at most 2^{18}, so all required roots of unity exist in $\mathbb{F}_\ell$.

For $6 \leq b \leq 73727$, the largest needed fundamental row is 73732 and the largest needed cotangent index is 73731. All recurrence denominators are invertible modulo ℓ. This is immediate for $2(2n+1)$ in (4.16); for the other diagonal use

$$d_s = 2\frac{(2s)!}{(2s-10)!}\frac{2s-5}{s}, \qquad s \geq 6. \tag{4.17}$$

Every displayed integer factor is positive and less than ℓ. Induction therefore identifies the computed residues with the reductions of the exact rational coefficients.

Proposition 4.4 (Finite gap-ten certificate)**.** *For every integer $6 \leq b \leq 73727$, the rational determinant $f_{b,5}$ has a nonzero reduction modulo* 998244353*.*

Exact computational proof. The program `verify_gap_ten.cpp` constructs the necessary cotangent coefficients and five fundamental columns using integer arithmetic only. It then evaluates all 73722 five-by-five determinants by Gaussian elimination over $\mathbb{F}_\ell$. Every residue is nonzero. The full indexed list is `gap_ten_certificate.csv`. Its SHA-256 identifier is

`f3f749a4c4e02c27b32a58e1f65714bab568a08f3b0ebdb0fd8cb8d4bf74184d`

The recurrence and the nonzero residues are the certificate; its hash serves only to identify the supplied record.

Every fundamental recurrence is independently checked after the causal computation by two full polynomial products per column. The cotangent coefficients are independently checked against (4.16), after their initial construction by formal division of cosine by normalized sine. The separate Python verifier, which does not call the NTT code, recomputes 506 boundary determinants with the direct quadratic recurrence, checks 64 cotangent coefficients against rational Bernoulli numbers, and compares the original Bernoulli matrix (4.5) and its full normalization (4.9) for every $6 \le b \le 18$. All these checks pass; their exact values are supplied in `gap_ten_independent_checks.json`. □

Propositions 4.3 and 4.4, followed by (4.9) and (4.7), prove Theorem 4.1. The finite-field part proves nonvanishing, not the real signs of the corresponding rational numbers. Indeed direct rational checks give $f_{6,5} > 0$, $f_{7,5} < 0$, and $f_{18,5} > 0$. A blanket positivity claim at every admissible index would be false.

4.7 Why the finite recurrence can be verified efficiently

The accelerated calculation has a short exact algebraic description. Define formal series

$$S(z) = \sum_{n\ge 0} \frac{(-1)^n z^n}{4^n (2n+1)!}, \qquad C(z) = \sum_{n\ge 0} \frac{(-1)^n z^n}{4^n (2n)!}.$$

Since $S(0) = 1$, its inverse exists over $\mathbb{F}_\ell$, and

$$1 - \tfrac{1}{2} \sum_{n\ge 1} \beta_n z^n = C(z)/S(z).$$

Newton iteration $U \mapsto U(2 - SU)$ doubles the number of correct inverse coefficients. The number-theoretic transform computes each polynomial product exactly in the finite field; it introduces no roundoff error.

For a fundamental column $r_u = R_{u,j}$, the recurrence is

$$d_s r_s = b_s \sum_{u<s} \beta_{s-u} r_u + \sum_{u<s} \beta_{s-u} a_u r_u.$$

Both sums are causal convolutions. Divide an index interval into two equal halves, compute its left half, add both left-to-right convolutions, and then compute its right half. Induction on the interval length proves that every already determined coefficient contributes exactly once to every later equation. The transforms for a given interval length use the same cotangent kernel and can be reused across all five columns.

In a cyclic transform of length L, the left input is supported in $[0, L/2)$ and the kernel in $[0, L)$. All wrapped terms land in $[0, L/2)$; none contaminates the right half being updated. This proves the validity of the shorter cyclic convolution used by the implementation. For a fixed half-gap m, the resulting cost is $O(mN \log^2 N + Nm^3)$ field operations and $O(mN)$ stored field elements, using ordinary cubic elimination for the consecutive boundary determinants. The same method applies to any prescribed half-gap; it is not tied to five columns.

5 Inverse harmonic coordinates from bounded-size determinants

For this section let $m \geq 1$ be arbitrary, $b > m$, and use the same definitions with 10 replaced by $2m$ and 5 by m. Thus $p = 2b - 2m$, $q = 2b$, $F_{b,m} = (R_{b+i,j})_{i,j=1}^{m}$, and assume $f_{b,m} = \det F_{b,m} \neq 0$. This assumption is now a theorem for every $b > 5$ when $m = 5$.

For an integer forcing index $N > m$, define the causal Green column $\mathsf{G}_{s,N}$ by zero initial values for $1 \leq s \leq m$ and

$$d_s \mathsf{G}_{s,N} = \delta_{sN} + \sum_{u<s} \beta_{s-u}(b_s + a_u)\mathsf{G}_{u,N} \qquad (s > m). \tag{5.1}$$

In particular $\mathsf{G}_{s,N} = 0$ for $s < N$ and $\mathsf{G}_{N,N} = 1/d_N$. Put

$$\mathbf{r}_n = (R_{n,1}, \ldots, R_{n,m}), \qquad \mathbf{g}_N = (\mathsf{G}_{b+1,N}, \ldots, \mathsf{G}_{b+m,N})^{\mathsf{T}}.$$

Theorem 5.1 (Every inverse coefficient in dimension $m+1$)**.** *For $0 \leq k, r < b$, set $n = b - k$ and $N = r + m + 1$. Then*

$$\boxed{(M_{b,m}^{-1})_{k,r} = \frac{2(-1)^{n+N}(2n)!}{(2r+1)!} \frac{\det \begin{pmatrix} F_{b,m} & \mathbf{g}_N \\ \mathbf{r}_n & \mathsf{G}_{n,N} \end{pmatrix}}{\det F_{b,m}}.} \tag{5.2}$$

The determinant in the numerator has order $m+1$, independent of b. Every entry is rational and specified by a finite recurrence.

Proof. Let $J_{m,b}$ be the coefficient matrix of the cotangent operator (4.10), with 10 replaced by $2m$, acting on $\sum_{n=1}^{b} r_n z^{2n}$ and with output rows $N = m+1, \ldots, b+m$. Its equation with unit forcing at N is the recurrence (5.1), with the boundary conditions $r_{b+1} = \cdots = r_{b+m} = 0$ imposed after truncation. The general full solution is

$$r_s = \mathbf{r}_s \mathbf{c} + \mathsf{G}_{s,N}.$$

Its m terminal conditions give $F_{b,m}\mathbf{c} = -\mathbf{g}_N$. Hence

$$(J_{m,b}^{-1})_{n-1,r} = \mathsf{G}_{n,N} - \mathbf{r}_n F_{b,m}^{-1} \mathbf{g}_N = \frac{\det \begin{pmatrix} F_{b,m} & \mathbf{g}_N \\ \mathbf{r}_n & \mathsf{G}_{n,N} \end{pmatrix}}{\det F_{b,m}}.$$

To restore the original scales, let $K_{m,b}$ be the analogous hyperbolic-cotangent coefficient matrix. If M^{rev} denotes column reversal, direct comparison with (4.5) gives

$$K_{m,b} = \mathrm{diag}_{0 \leq r < b}\left(\frac{2}{(2r+1)!}\right) M_{b,m}^{\mathrm{rev}} \, \mathrm{diag}_{1 \leq n \leq b}((2n)!).$$

Since $2B_{2h}/(2h)! = (-1)^{h-1}\beta_h$ for $h \geq 1$, its entries satisfy $(K_{m,b})_{r,n-1} = (-1)^{N+n}(J_{m,b})_{r,n-1}$. The same row and column signs therefore occur in the inverse. Finally the original row k corresponds to the reversed index $n = b - k$. These factors give precisely (5.2). □

This formula constructs each inverse row without expanding a b-square determinant. It does not remove the need to compute the finite Green column, and it makes no minimality claim about the determinant size. The independent verifier checks all 293 entries of the resulting inverse matrices at $b = 6, 7, 8, 12$ directly against the original source matrices, and separately checks an actual six-by-six bordered determinant.

6 Ordinary polygamma identities at every gap of ten

Let $b \geq 6$, $p = 2b - 10$, $q = 2b$, and let $\lambda_{b+r} = (M_{b,5}^{-1})_{0,r}$ for $0 \leq r < b$. Define

$$K_b(y) = \sum_{j=b}^{2b-1} \lambda_j y^{2j+1},$$

$$P_b(x) = K_b(x + \tfrac{1}{2}) - K_b(x - \tfrac{1}{2}) = \sum_{r=0}^{q-1} w_r x^{2r}. \tag{6.1}$$

By Theorem 5.1, all its coefficients can be given by six-by-six determinants and the five-column recurrence. With $c_{v,h}$ from (4.1), put

$$u_h = \sum_{e=0}^{h} c_{p,e} c_{q,h-e} \quad (0 \leq h \leq 5), \qquad S_b(x) = \sum_{h=0}^{5} \left(\sum_{e=0}^{h} u_e w_{q-1-h+e} \right) x^{10-2h}.$$

Corollary 6.1 (An all-order rational family)**.** *For every integer $b \geq 6$,*

$$\boxed{\sum_{n=0}^{\infty} \left\{ \frac{P_b(n+1/2)}{(p-1)!(q-1)!} \psi^{(p-1)}(n+1)\psi^{(q-1)}(n+1) - S_b(n+1/2) \right\} = -\frac{1}{2}.} \tag{6.2}$$

The sum is absolutely convergent. The polynomial P_b has exact degree $2q-2$; its leading coefficient is

$$[x^{2q-2}]P_b(x) = \frac{4b(4b-1)f_{b-1,5}}{d_{b+5}f_{b,5}} \neq 0, \tag{6.3}$$

where at the initial index one uses the auxiliary recurrence value

$$f_{5,5} = \frac{1}{3636981591775743187067470801343195813555208192000000000000}. \tag{6.4}$$

Proof. The inverse row gives $\sum_{j=b}^{2b-1} \lambda_j C_j = Z_0 = -1/2$. Expanding the centered difference in (6.1) and using (4.2) gives $\sum_r w_r D_r = -1/2$. By (4.1), the nonnegative powers in $P_b(x_n)A_p(n)A_q(n)$ are exactly $S_b(x_n)$, and the remaining summand is $O(x_n^{-2})$. The continued sums of the subtraction monomials vanish because $\zeta(-2h, 1/2) = 0$ for every $h \geq 0$. Equivalently, near spectral parameter zero, subtracting this finite polynomial gives a normally convergent Dirichlet series and leaves only those Hurwitz-zeta terms. Its value at zero is the ordinary sum of the subtracted terms. Finally $A_v(n) = \psi^{(v-1)}(n+1)/(v-1)!$ for positive even v.

For $b \geq 7$, delete the last row and first column of $M_{b,5}$. Reindexing the two orders down by two and the two source indices down by one identifies the minor exactly with $M_{b-1,5}$, including the omitted-binomial convention. Cramer's rule and (4.9) give

$$\lambda_{2b-1} = (-1)^{b-1} \frac{\det M_{b-1,5}}{\det M_{b,5}} = \frac{4bf_{b-1,5}}{d_{b+5}f_{b,5}}.$$

The centered difference multiplies this highest coefficient by $4b-1$, proving (6.3). At $b = 6$ direct rational recurrence gives (6.4), and the original matrix gives $\lambda_{11} = 132/127$, verifying the same formula without introducing a zero-order tail. Nonvanishing follows from Theorem 4.1 and the explicit auxiliary value. □

6.1 The newly covered pair $(14, 24)$

Take $b = 12$, so $p = 14$ and $q = 24$. Formula (6.2) is then the explicit ordinary identity

$$\sum_{n\geq 0}\left\{\frac{P_{12}(n+1/2)}{13!\,23!}\psi^{(13)}(n+1)\psi^{(23)}(n+1) - S_{12}(n+1/2)\right\} = -\frac{1}{2},$$

with $\deg P_{12} = 46$ and $\deg S_{12} = 10$. Its leading coefficients are

$$[x^{46}]P_{12}(x) = \frac{19666726079302752587487468896 0}{14009065684434488629739697248281},$$
$$[x^{10}]S_{12}(x) = \frac{19666726079302752587487468896 0}{41887106396459121002921694772360 19}.$$

The second denominator is $13 \cdot 23$ times the first. The file `gap_ten_14_24_example.json` contains all 24 even coefficients of P_{12}, all six coefficients of S_{12}, and the twelve coefficients of K_{12} as exact rationals. The formula (5.2) is an alternative finite specification of the same coefficients. The generator checks the inverse against the original Bernoulli matrix and the entire subtraction polynomial symbolically. This pair lies beyond the previous smaller-order-through-twelve coverage.

More generally, replace row zero by row k of the inverse in (6.1). The identical construction, with the same degree bound and subtraction rule, gives the right-hand side $\zeta(2k)$ for every $0 \leq k < b$. Thus the completed reconstruction theorem gives a family of ordinary convergent polygamma-product representations for every even zeta coordinate in the source block.

7 Closed Gamma primitive certificates and analytic infinite jets

The newer of the two incoming *Universal Gamma Normal Forms* packages gives every canonical coefficient of the inverse-Gamma differential hierarchy. Its research questions 1 and 6 ask, respectively, for a nonrecursive formula for the compensating total derivative and for analytic meaning of the infinite-jet generator. We answer those questions here. The algebraic device used for the first answer is the classical one-dimensional Euler homotopy operator [15, 16]. Its application to the incoming coefficient theorem, rather than the homotopy operator itself, is the continuation result. The convergence theorem below supplies an analytic domain for actual holomorphic logarithmic derivatives.

7.1 Definitions and the inherited coefficient formula

Let

$$\mathcal{R} = \mathbb{Q}[u_0, u_1, \ldots], \qquad Du_j = u_{j+1}, \qquad \mathrm{wt}(u_j) = j + 1.$$

The factor degree of $u^{\boldsymbol{a}} = \prod_j u_j^{a_j}$ is $|\boldsymbol{a}| = \sum_j a_j$. A monomial is canonical if it is a power of u_0, or if its largest positive index occurs at least twice. The canonical complement of $D\mathcal{R}$ gives unique polynomials $\mathcal{N}(P), \mathcal{H}(P)$ satisfying

$$P = \mathcal{N}(P) + D\mathcal{H}(P), \qquad [1]\mathcal{H}(P) = 0. \tag{7.1}$$

This complement is classical [17, Proposition 4.10]. For completeness, a unique largest derivative u_j can be removed by writing a suitable multiple of the monomial with u_j replaced by u_{j-1} as a total derivative. The remainder has lower highest index. This proves existence by termination. If H is nonconstant with highest index j, then DH contains the nonzero part $u_{j+1}\partial H/\partial u_j$, whose monomials are noncanonical. Thus a canonical total derivative is zero, proving uniqueness.

Define

$$A_0 = 1, \qquad A_{m+1} = u_0 A_m - D A_m, \qquad N_m = \mathcal{N}(A_m), \quad H_m = \mathcal{H}(A_m). \tag{7.2}$$

Their raw generating series is

$$\sum_{m\ge 0} A_m \frac{t^m}{m!} = \exp\left(\sum_{j\ge 0} \frac{(-1)^j u_j t^{j+1}}{(j+1)!}\right). \tag{7.3}$$

Consequently, at weight $m = \sum (j+1)a_j$,

$$a(\boldsymbol{a}) := [u^{\boldsymbol{a}}]A_m = \frac{(-1)^{m-|\boldsymbol{a}|} m!}{\prod_j \bigl(a_j!\,((j+1)!)^{a_j}\bigr)}. \tag{7.4}$$

Here is the precise part of the incoming theorem used below [6]. Set

$$C_j(v) = \frac{(v+1)^{j+1} - (v-1)^{j+1}}{2^{j+1}(j+1)!}. \tag{7.5}$$

The coefficient $n(\boldsymbol{a}) = [u^{\boldsymbol{a}}]N_m$ is one for u_0^m, zero for a unique highest positive derivative, and zero when $\sum j a_j$ is odd. In the remaining case, with highest index $d \ge 1$ and $a_d \ge 2$, it is

$$n(\boldsymbol{a}) = \frac{(-1)^d m!}{4^d d!(d+1)! \prod_j a_j!} \int_0^1 (1-v^2)^d C_d(v)^{a_d - 2} \prod_{j=1}^{d-1} C_j(v)^{a_j}\, dv. \tag{7.6}$$

The integrand is a polynomial, so this is an explicit finite rational number. We retain attribution for this coefficient formula; the new primitive formula starts with it as an input.

7.2 A nonrecursive formula for every compensator

For a differential polynomial define

$$E(P) = \sum_{j\ge 0} (-D)^j \frac{\partial P}{\partial u_j}, \tag{7.7}$$

$$B(P) = \sum_{j\ge 1} \sum_{k=0}^{j-1} u_{j-1-k} (-D)^k \frac{\partial P}{\partial u_j}. \tag{7.8}$$

Let K denote the factor-degree operator: $K(u^{\boldsymbol{a}}) = |\boldsymbol{a}| u^{\boldsymbol{a}}$. Its inverse is defined on polynomials without constant term by division of each coefficient by its positive factor degree.

Lemma 7.1 (Euler homotopy identity)**.** *For every $P \in \mathcal{R}$,*

$$KP = u_0 E(P) + D B(P). \tag{7.9}$$

If P has zero constant term and $E(P) = 0$, then

$$D\bigl(BK^{-1}P\bigr) = P. \tag{7.10}$$

Proof. For an arbitrary polynomial Q and $j \ge 1$, direct differentiation of a telescoping sum gives

$$u_j Q - u_0 (-D)^j Q = D\left(\sum_{k=0}^{j-1} u_{j-1-k} (-D)^k Q\right).$$

Substitute $Q = \partial P/\partial u_j$ and sum over j, including the $j = 0$ term separately. Euler's identity for factor degree yields (7.9). The operators D and B preserve factor degree, while E lowers it by one. Apply the identity separately to each positive factor-degree component of P, then divide by that degree. This proves (7.10). In particular no analytic integration and no recursive integration by parts is involved. □

Theorem 7.2 (Complete finite primitive formula)**.** *For every $m \geq 1$,*

$$\boxed{H_m = BK^{-1}(A_m - N_m).} \tag{7.11}$$

Equivalently, let $p(\boldsymbol{a}) = a(\boldsymbol{a}) - n(\boldsymbol{a})$ with the explicit coefficients in (7.4)–(7.6). Then

$$\boxed{H_m = \sum_{\sum(j+1)a_j = m} \frac{p(\boldsymbol{a})}{|\boldsymbol{a}|} \sum_{j\geq 1} a_j \sum_{k=0}^{j-1} (-1)^k u_{j-1-k} D^k u^{\boldsymbol{a}-\boldsymbol{e}_j}.} \tag{7.12}$$

All sums are finite. Together with (7.6), this determines both polynomials in the canonical decomposition without any recursive normal reduction.

Proof. The polynomial $A_m - N_m$ is a total derivative by (7.1). The identity $E(DQ) = 0$ follows by cancellation from $\partial_{u_j} D = D\partial_{u_j} + \partial_{u_{j-1}}$. Lemma 7.1 therefore shows that the right side of (7.11) differentiates to $A_m - N_m$. It has zero constant term, so uniqueness in (7.1) identifies it with H_m. Expanding the two operators gives (7.12). □

If a formula without powers of D is desired, every term is expanded by

$$D^k u^{\boldsymbol{b}} = k! \sum_{\substack{\sum_{r\geq 0} b_{i,r} = b_i \ (i\geq 0) \\ \sum_{i,r} r b_{i,r} = k}} \prod_i \frac{b_i!}{\prod_r b_{i,r}! (r!)^{b_{i,r}}} \prod_{i,r} u_{i+r}^{b_{i,r}}. \tag{7.13}$$

This is simply the multinomial Leibniz rule. Thus the word "nonrecursive" describes a literal finite coefficient formula, not an instruction to run an integration-by-parts recurrence in disguise.

7.3 An explicit higher Gamma primitive

For example,

$$N_6 = u_0^6 - 15u_0^2u_1^2 + u_2^2, \tag{7.14}$$

$$H_6 = -3u_0^5 + 20u_0^3u_1 - 15u_0^2u_2 - 15u_0u_1^2 + 6u_0u_3 + 9u_1u_2 - u_4. \tag{7.15}$$

At weight eight the normal part is

$$N_8 = u_0^8 - 70u_0^4u_1^2 + 28u_0^2u_2^2 - 7u_1^4 - u_3^2, \tag{7.16}$$

and the companion polynomial, supplied completely by the same formula, is

$$\begin{aligned} H_8 = &-4u_0^7 + 56u_0^5u_1 - 70u_0^4u_2 - 140u_0^3u_1^2 + 56u_0^3u_3 \\ &+ 252u_0^2u_1u_2 - 28u_0^2u_4 + 112u_0u_1^3 - 112u_0u_1u_3 \\ &- 84u_0u_2^2 + 8u_0u_5 - 98u_1^2u_2 + 20u_1u_4 + 36u_2u_3 - u_6. \end{aligned} \tag{7.17}$$

These low-weight examples are illustrations of the closed construction, not claims of new historical evaluations.

For a smooth function U, substitute $u_j = U^{(j)}$ and let $Y' = -UY$. The inherited weighted identity now has a fully nonrecursive right side:

$$\boxed{\int Y\big(N_m(U) + UH_m(U)\big)\,dx = -Y\big(A_{m-1}(U) + H_m(U)\big) + C.} \tag{7.18}$$

Indeed $(YA_{m-1})' = -YA_m$ and $(YH_m)' = Y(DH_m - UH_m)$. The formula follows on adding these two identities. The compensator UH_m and the endpoint value of YH_m are part of the formula; neither can be discarded.

Taking $U = \psi$ and $Y = 1/\Gamma$ gives exact Gamma–polygamma antiderivatives at every weight. Taking instead

$$F_q(x) = \frac{(-1)^q}{q+1}\zeta^{(q+1)}(0,x), \quad U_q(x) = -\gamma_q(x), \quad Y_q(x) = e^{-F_q(x)} \tag{7.19}$$

gives the corresponding all-index Stieltjes primitives. Here $\zeta^{(q+1)}$ denotes an order derivative, not an argument derivative. The standard identity $\partial_x\zeta(s,x) = -s\zeta(s+1,x)$ and the Laurent convention $\zeta(1+s,x) = s^{-1} + \sum_{q\geq 0}(-1)^q\gamma_q(x)s^q/q!$ give $F_q' = -\gamma_q = U_q$ [9]. Thus $Y_q' = -U_qY_q$, as required. At $q = 0$, $F_0 = \log\Gamma - \frac{1}{2}\log(2\pi)$.

7.4 The infinite-jet formula is analytic for holomorphic fields

The incoming coefficient theorem is formal in infinitely many jets. The following result permits their actual substitution from a holomorphic function. Define the entire function

$$\phi(z) = \frac{e^z - 1 - z}{z^2}, \qquad \phi(0) = \frac{1}{2}.$$

For x in a domain of holomorphy of U, put $x_j = t^{j+1}U^{(j)}(x)$ and $S_{r-1}(v) = \sum_{j=1}^{r-1} C_j(v)x_j$.

Theorem 7.3 (Analytic normal generator)**.** *Suppose U is holomorphic on $|x - x_0| < R$. The series*

$$\boxed{\mathfrak{N}_U(x,t) = e^{tU(x)}\left[1 + \frac{1}{2}\sum_{r\geq 1}\frac{(-1)^r x_r^2}{4^r r!(r+1)!}\int_{-1}^{1}(1-v^2)^r e^{S_{r-1}(v)}\phi(C_r(v)x_r)\,dv\right]} \tag{7.20}$$

converges absolutely and locally uniformly when

$$|x - x_0| + |t| < R. \tag{7.21}$$

It is holomorphic there and has Taylor series

$$\mathfrak{N}_U(x,t) = \sum_{m\geq 0} N_m(U(x))\frac{t^m}{m!}. \tag{7.22}$$

Proof. Cover any compact subset of (7.21) by finitely many sufficiently small compact polydisks. On each such polydisk choose ρ with $\sup|t| < \rho < R - \sup|x - x_0|$. Cauchy's estimate supplies $M > 0$ such that, uniformly there,

$$|U^{(r)}(x)| \leq Mr!\rho^{-r}, \qquad |t| < \rho.$$

For real $-1 \leq v \leq 1$, the explicit positive-coefficient formula for C_j gives $|C_j(v)| \leq 1/(j+1)!$. Hence

$$\sum_{j\geq 1}|C_j(v)x_j| \leq M|t|\sum_{j\geq 1}\frac{(|t|/\rho)^j}{j+1} < \infty$$

uniformly. Also $|\phi(z)| \leq e^{|z|}/2$, from its elementary integral representation $\phi(z) = \int_0^1(1-s)e^{sz}\,ds$. Apart from a uniform constant, the absolute value of the rth summand of (7.20) is consequently at most

$$\frac{|x_r|^2}{4^r r!(r+1)!} \leq \frac{M^2|t|^2}{r+1}\left(\frac{|t|}{2\rho}\right)^{2r}.$$

This is summable. The same reasoning applies on each compact subset, proving normal convergence and holomorphy. At any fixed power of t only finitely many r contribute. Expanding the exponential and ϕ, then using (7.6), gives exactly the canonical coefficients. This proves (7.22). □

The factor ϕ removes all apparent singularities before any infinite summation. In particular odd-index kernels cause no finite-part integral in the analytic formula.

Theorem 7.4 (Analytic primitive and translation identity)**.** *Under the same hypothesis, the canonical series*

$$\mathfrak{H}_U(x,t) = \sum_{m\geq 1} H_m(U(x))\frac{t^m}{m!} \tag{7.23}$$

converges locally uniformly on the smaller domain

$$|x - x_0| + 2|t| < R. \tag{7.24}$$

If $F' = U$ locally, then on this domain

$$\boxed{e^{F(x)-F(x-t)} = \mathfrak{N}_U(x,t) + \partial_x \mathfrak{H}_U(x,t).} \tag{7.25}$$

The expression on the left is independent of the additive constant in F.

Proof. We provide the convergence justification, since the formal decomposition alone is insufficient. Let $P = A - N$ denote the formal generating-series difference. The coefficientwise homotopy formula can be written as

$$\mathfrak{H}_U = \int_0^1 \sum_{j\geq 1}\sum_{k=0}^{j-1} U^{(j-1-k)}(x)(-\partial_x)^k \left(\partial_{u_j} P\right)\Big|_{u_r=\lambda U^{(r)}(x)} d\lambda. \tag{7.26}$$

The factor of λ from differentiating $P(\lambda u)$ cancels the $1/\lambda$ in $K^{-1}P = \int_0^1 P(\lambda u)d\lambda/\lambda$. There is therefore no singular endpoint in (7.26).

Fix a compact subset of (7.24). Around each of its x values choose a disk of radius $\rho > |t|$ whose closure still satisfies the normal-generator condition with the same t. The raw exponential gives immediately

$$|\partial_{u_j} A| \leq C\frac{|t|^{j+1}}{(j+1)!}.$$

Differentiation of (7.20), uniformly for $0 \leq \lambda \leq 1$, gives the same bound for $\partial_{u_j} N$. To see the only potentially different term, differentiate the $r = j$ summand in its nonsingular form and use

$$\left|\frac{d}{dy}\{y^2\phi(cy)\}\right| = \left|y\frac{e^{cy}-1}{cy}\right| \leq |y|e^{|cy|},$$

with continuous interpretation at $cy = 0$. The prefactor $1/(4^j j!(j+1)!)$ and Cauchy's jet estimate then give the required bound. For $r > j$ there is just the factor $t^{j+1}C_j$, and the already summable majorant applies. These bounds hold on a slightly larger compact set, so Cauchy's estimate in x yields

$$\left|\partial_x^k(\partial_{u_j} P)(\lambda U)\right| \leq C\frac{k!\,|t|^{j+1}}{(j+1)!\rho^k}, \qquad |U^{(j-1-k)}(x)| \leq M\frac{(j-1-k)!}{\rho^{j-1-k}}.$$

If necessary shrink ρ slightly to have both bounds with one common radius still exceeding $|t|$. Since

$$\sum_{k=0}^{j-1}\frac{k!(j-1-k)!}{(j+1)!} \leq \frac{1}{j+1},$$

the jth group in (7.26) is bounded by $CM|t|^{j+1}\rho^{1-j}/(j+1)$, a summable series. The same bounds prove normal convergence on compact subsets. Coefficient extraction is justified and reproduces Theorem 7.2. Finally the convergent raw series is $\exp(\int_{x-t}^{x} U(s)ds) = e^{F(x)-F(x-t)}$. The coefficientwise equality $A_m = N_m + DH_m$, now between holomorphic functions, proves (7.25). □

Corollary 7.5 (Gamma and Stieltjes translation formulas)**.** *For positive x and $|t| < x/2$,*

$$\frac{\Gamma(x)}{\Gamma(x-t)} = \mathfrak{N}_{\psi}(x,t) + \partial_x \mathfrak{H}_{\psi}(x,t), \tag{7.27}$$

$$\exp(F_q(x) - F_q(x-t)) = \mathfrak{N}_{-\gamma_q}(x,t) + \partial_x \mathfrak{H}_{-\gamma_q}(x,t), \tag{7.28}$$

with F_q as in (7.19). These are ordinary holomorphic identities in t and local holomorphic identities in x. On a positive interval $[a,b]$ with $|t| < a/2$ they may be integrated termwise and retain the exact endpoint difference $\mathfrak{H}_U(b,t) - \mathfrak{H}_U(a,t)$.

Proof. Both ψ and γ_q are holomorphic on the right half-plane. At a fixed positive x, every disk of radius $R < x$ is available. Take $2|t| < R < x$ in Theorem 7.4, then substitute the standard primitives. Compactness gives uniformity on $[a,b]$. □

No sharp convergence radius is asserted. The purpose of the domain is to justify infinite-jet substitution, coefficientwise differentiation, and anchored integration for actual special functions.

7.5 Independent finite checks and the remaining question

The accompanying `normal_forms.py` constructs A_m by its differential recurrence, constructs N_m independently from the rational beta coefficient formula, and computes H_m in two ways: the direct canonical reduction and the closed Euler homotopy. Through weight 20 all results agree, including $A_m = N_m + DH_m$ and the full Euler–boundary identity. There are 106 recorded exact assertions, including a rejected coefficient corruption. These checks support implementation correctness; the proofs of the all-order assertions are the theorems above.

An additional exploratory calculation through weight 24 suggests that for $2 \le m \le 24$ every monomial of weight $m-1$ occurs in H_m, with sign $(-1)^{m-|\boldsymbol{a}|}$ for its coefficient. We do not use or claim this as a theorem. A positivity formula for the absolute coefficients in (7.12) would strengthen the closed primitive result and explain the observed full support.

8 Source audit and proposed integration corrections

The audit found no new counterexample to a theorem in the inherited material on which the proofs here depend. It did find several places where the status of a direction must be updated when the newer reports are integrated. An older question remains a valid historical record even when a later incoming article answers it. The appropriate change is a dated completion note and a link to the proof, while retaining the original provenance.

A verified editorial duplication. The retrieved manuscript `README.md` repeats the paragraph beginning “The complete 1,032-input first-letter X family consecutively. Remove its second occurrence while retaining the first and the surrounding research-status text. This duplication was already noted in the latest incoming source-status report; it is an editorial cleanup, not a new mathematical erratum. The separate statements about pending canonical analytic integration remain appropriate and should not be rewritten as though every incoming result had already been incorporated.

8.1 What should change in the current research map

The fixed-gap map should now record complete gaps $2, 4, 6, 8, 10$. The gap-eight proof belongs to [4]; only gap ten is completed in this article. The smaller-order-through-twelve results from earlier work are another source of coverage and should not be counted again as new gap-ten cases. The pair $(14, 24)$ is included as an explicit example outside that earlier smaller-order range.

The compact inverse-row question receives a constructive bounded-size answer in Theorem 5.1: at fixed half-gap m, every entry is a quotient involving an $(m+1)$-square determinant. This does not answer the stronger request for a short product or for a recurrence using only a fixed number of final rows. Nor does the gap-ten proof remove the finite computational component; the source's request for a structural proof without finite completion remains open.

The three-cluster question B^r, C, D in *Gamma Normalizations and Colored Moments* is answered by the explicit dictionary in Section 3 [5]. A status note should retain the distinction between the finite colored dictionary for one spectral term and the convergent infinite spectral sum for the Gamma face. The much broader cluster pattern B^r, C^s, D^t with arbitrary multiplicities remains open in the form posed there.

The collision question should now distinguish three levels. The two-factor theorem for all positive monomial scales and all Stieltjes indices is inherited. The present arbitrary-factor theorem evaluates the complete local correction at Stieltjes index zero, uniformly for all positive paths. Complete scalar values are additionally obtained for the one-derivative centered family, including the uncentered triple. The corresponding arbitrary-factor theorem at positive Stieltjes indices is a further question.

Finally, the first and sixth research questions of the expanded *Universal Gamma Normal Forms* package receive the answers in Section 7. The universal normal coefficients themselves remain an attributed input. The new finite formula is for the compensating polynomial H_m, and the new analytic result is the convergence of the infinite substitutions for a holomorphic field.

8.2 Conventions that materially affect correctness

Nonvanishing does not imply positivity. The real fundamental determinant has $f_{6,5} > 0$, $f_{b,5} < 0$ for $7 \le b \le 17$, and $f_{18,5} > 0$. Thus the conjectural strengthening $f_{b,5} > 0$ at every admissible index is false. This is an explicit obstruction to that possible heuristic, not a claim that the predecessor asserted it. The finite-field certificate proves nonvanishing. Real positivity is proved separately only for $b \ge 73728$.

The constant depends on the scale relation. The nonresonant triple in (1.1) cannot be applied to $(\alpha_1, \alpha_2, \alpha_3) = (3, 2, 1)$. The extra terms in (2.21) are a concrete correction to such an extension. A power–logarithm constant is also different from the ordinary limit of the original divergent family. The explicit Q subtraction states exactly which ordinary limit is available.

Exact regular coefficients are necessary for arbitrary paths. Replacing $A_{S,d}(\boldsymbol{a})$ by a prematurely truncated expression in the shifts can destroy uniformity. For example, if $a_{\min} = e^{-1/t^2}$ and $a_{\max} = t$, then $a_{\max}|\log a_{\min}| = 1/t$ does not tend to zero. Theorem 2.2 keeps the shift dependence of every retained regular coefficient exactly; its remainder proof therefore needs no assumption on relative rates.

Labelled derivatives and confluent derivatives differ. The mixed operator $\partial_{C_1} \cdots \partial_{C_m}$ is applied before $C_1 = \cdots = C_m$ is imposed. A total derivative along that diagonal has extra multiplicities and other partial derivatives. The factorial $(m-1)!$ in the confluent divided

difference is part of the theorem. Likewise, the three-cluster primitive must retain its two edge terms; its mixed derivative alone does not determine the primitive without boundary data.

Endpoints and branches stay in the formulas. The compensator UH_m and the endpoint term YH_m in (7.18) are essential. Discarding a total derivative under an unweighted finite-part integral requires its own endpoint calculation. For the colored formulas the displayed real branch is continued using letters outside the integration path. Iterated-integral shuffle keeps those letters; a stuffle conversion outside the initial disk needs additional branch bookkeeping if it introduces new cumulative colors.

The arithmetic questions remain arithmetic questions. An invertible rational coefficient matrix does not prove independence of its evaluated constants. A depth-three representation does not prove that depth three is necessary. The Gaussian S_6 identity and the current revised S_8 period candidate remain unresolved in the inspected research programme; the present results provide no proof of those equalities. These restrictions should remain visible in any combined manuscript overview.

9 Further research questions

The following questions are chosen for their potential to produce identities, structural formulas and exact evaluations. The thresholds used in the gap-ten proof are tools for nonvanishing, rather than the main objects of interest.

Q1. Arbitrary unequal even harmonic tails. Prove or disprove $\det M_{b,m} \neq 0$ for every $b > m \geq 1$. Can a transform, a sign-regular kernel, a Padé interpretation or a spectral argument account for the nonvanishing despite the finite sign changes? The full two-parameter problem is stronger than settling each of finitely many gaps.

Q2. The next fixed gap and a reusable exact completion theorem. Complete gap twelve using the effective Gamma basis and the causal convolution method. A useful further result would be an explicit algorithm that takes m, derives a valid analytic threshold and produces a fully replayable finite certificate, including a principled choice of primes when one reduction vanishes. A zero residue is not evidence of a zero rational determinant.

Q3. Closed Green columns and inverse harmonic coordinates. Find a hypergeometric, Padé or integrable representation of $\mathsf{G}_{s,N}$ that eliminates its long causal recurrence. Determine whether the bounded determinant in (5.2) admits a short quotient of special functions. The resulting formulas would turn high-order zeta-coordinate sums into more recognizable identities.

Q4. Positive-index Stieltjes collisions with three or more factors. For $q_i \geq 0$, extend the uniform local theorem to $\prod_i \partial_x^{r_i} \gamma_{q_i}(x + a_i)$, with a fully evaluated correction and a uniform error for arbitrary positive paths. The singular kernels now contain logarithms of several shifted variables. An effective route is to introduce independent spectral powers before differentiating, while retaining every regular coefficient needed for uniformity. A formal derivative of an unproved uniform family is insufficient.

Q5. A finite classification of resonant scale contributions. For a fixed derivative pattern and fixed number of factors, classify the integer relations among the α_i that can contribute to the log-free constant. The rational kernel makes each specified monomial case finite. Is there a compact chamber description with explicit transition polynomials and amplitude logarithms? The difference between (1.1) and (2.21) provides a first exact test case.

Q6. More than one derivative-bearing factor. Find full scalar evaluations for centered products with two or more nonzero derivative orders. The local correction is already given by Theorem 2.2; the unresolved step is the merged product. Which patterns are total derivatives of finite polygamma polynomials, and which force marked-zeta or genuine new moment coordinates?

Q7. Generating functions for the ordered constant rows. Use (2.14) to sum in N and in the derivative position r. Seek polylogarithmic or Gamma expressions for these double generating functions, and relations with harmonic-number weights. The adjacent-difference and zero-sum identities give two exact constraints on any proposed answer.

Q8. A finite primitive for more displaced masses. Integrate (3.21) for $B^r, C_1, \dots, C_m$ with $m \geq 3$, keeping all lower-dimensional face data. Is depth at most $m+1$ sufficient uniformly in r? Can the edge reconstruction be expressed by a finite symmetric dictionary with controlled letters, rather than by unresolved parameter derivatives?

Q9. Three clusters with arbitrary multiplicities. Extend B^r, C, D to B^r, C^s, D^t and identify the smallest useful uniform depth bound. The one-unit multiplicities in the present proof produce an elementary simplex resolvent; the general problem requires retaining additional Dirichlet powers without losing the finite colored description.

Q10. Arithmetic collapse of the Gamma spectral sum. Find special positive a and nonnegative masses for which (3.19) reduces to finitely many Gamma, zeta or multiple-polylogarithm constants. Coincident masses, rational shifts and imaginary-Gamma multiplication identities give natural families to test. Any reduction should state its period basis and branch conventions before invoking a numerical relation search.

Q11. Signs and full support of the compensators. The exact data through weight 24 suggest that for every $m \geq 2$ all monomials of weight $m-1$ occur in H_m, with coefficient sign $(-1)^{m-|\boldsymbol{a}|}$. Prove or refute this statement. A positive integral formula for the absolute coefficients would explain the cancellations in the Euler-homotopy expression and could also control their denominators.

Q12. Sharp analytic domains and continuation of the canonical series. Determine the true convergence domains of $\mathfrak{N}_U$ and $\mathfrak{H}_U$ for $U = \psi$ and $U = -\gamma_q$. The sufficient domain $|t| < x/2$ makes the translation identity rigorous; it is not asserted to be sharp. Can the normal and primitive pieces continue separately to the natural singular locus of the Gamma or Stieltjes ratio, or do they develop additional singularities that cancel only in their sum?

Q13. Label-preserving and difference-algebra normal forms. Develop a canonical complement for several independent fields that retains the individual Gamma factors and their mixed derivative relations. In a separate direction, replace the ordinary derivation by a difference operator and derive the changed homotopy and product rules. Simply substituting a q-Gamma function into an ordinary derivative formula would not resolve this structural question.

Q14. The unresolved Gaussian periods. Continue the Gaussian S_6 and revised S_8 problems in a declared relation system. Seek a new analytic transformation, a complete iterated-integral reduction or a certified functional identity. Finite relation searches and numerical agreement can guide that work but do not become proofs merely by increasing precision or the size of a relation matrix.

10 Proof certificates and reproducibility

The archive contains the standalone source and PDF of this article, the same text in modular sections for integration, exact coefficient data, numerical receipts, and the programs used to generate them. The build is local and requires no network access. The scripts never need repository credentials or external services.

10.1 What was checked

Component	Recorded check	Mathematical role
Gap-ten finite interval	73722 nonzero residues for every $6 \le b \le 73727$ modulo 998244353.	Exact finite part of the proof.
Gap-ten independent audit	All full recurrence identities; 506 direct boundary determinants; 64 Bernoulli coefficients; original matrices for $6 \le b \le 18$; 293 inverse entries at $b = 6, 7, 8, 12$.	Independent normalization and implementation checks.
Gap-ten infinite tail	A complex Stirling remainder and a rational determinant/permanent margin for $b \ge 73728$.	Analytic proof, not a numerical extrapolation.
Polygamma contacts	21 exact kernel checks, including repeated nodes; the complete resonant triple; centered rows $N = 2, \dots, 8$, their adjacent differences and zero sums.	Exact finite checks of algebra and implementation.
Shifted-integral diagnostics	12 quadratures at 65 decimal working precision, with all three derivative positions and both centered and uncentered triples.	Asymptotic diagnostics; not interval certificates.
Three-cluster formulas	192 exact harmonic checks, 14 scalar comparisons, 24 mixed-derivative comparisons and one Gamma spectral comparison: 231 assertions in all.	Exact arithmetic plus floating-point diagnostics, separated in the receipt.
Gamma normal forms	106 exact assertions through weight 20, using an independent reducer, the beta coefficient formula and Euler homotopy, including a rejected coefficient corruption.	Finite verification of the all-order algebraic formulas.

The finite-field proof is reproducible from its recurrence and source code. Each transform is arithmetic in a prime field, with the prime, primitive root and nonzero denominators checked. No floating-point operation decides the nonvanishing statement. The certificate hash in Section 4 identifies the record, while the verifier establishes the mathematical content.

The three-cluster numerical comparisons use independently arranged formulas: cumulative-color sums, an ordinary scalar logarithmic integral and coefficients from the Dirichlet moment expansion. At 75 decimal working precision the scalar scaled residuals are below 1.5×10^{-75} and the mixed-derivative residuals below 6×10^{-72} for the recorded cases. These statements describe observed residuals, not certified errors in every reported digit.

For the collision quadratures, the largest residual after subtracting the exact contact and the merged baseline decreases from approximately 0.0187914 at $\epsilon = 10^{-3}$ to 0.000372220 at $\epsilon = 10^{-5}$. The proof of the limit is uniform dominated convergence. The quadratures are useful precisely because they compare the original positive-shift integral to the independently assembled finite correction.

10.2 Programs and exact records

The main reproduction command is

```
python3 reproduce.py --exact
```

It rebuilds the C++ verifier, recomputes the full gap-ten interval, compares its bytes with the included certificate, and runs the independent rational, contact and normal-form checks. The optional numerical suite is

```
python3 reproduce.py --numeric
```

The PDF is built from the standalone source with

```
latexmk -pdf Higher_Identities.tex
```

The file `README.txt` gives the dependencies and individual commands. Exact rational data use numerator/denominator strings; they do not require a computer algebra system to parse. The full $(14, 24)$ polynomial pair is in `code/gap_ten/gap_ten_14_24_example.json`.

10.3 Provenance and limits of the verification

The supplied source manifest records the bytes retrieved from the moving main branch. The repository HEAD reported during retrieval was

`e4ea0a4c7937d7ac0091048065222d1b58370482.`

The individual file hashes, rather than an assumption that every request was an atomic checkout of that HEAD, are the definitive identifiers of the inspected snapshots. The two similarly named *Universal Gamma Normal Forms* archives are both recorded; the expanded version with the suffix `(2)` supplies the two explicit research questions addressed here.

The principal analytic dependencies have been restated with their normalizations. The complementary-minor and Gamma-basis constructions in the gap-ten section retain their attribution to the ordered-boundary report; the universal normal coefficients retain theirs to the universal-normal-form report. Classical sources support the standard Gamma, Hurwitz-zeta, divided-difference and homotopy facts. A separate `INTEGRATION_NOTES.txt` maps the new conclusions to the existing questions without rewriting historical archives.

The proofs have been checked across the independent research sections, and the displayed finite formulas have the recorded computational support. This is a research contribution for further expert review and integration, not a claim that the entire surrounding manuscript or the worldwide literature has been exhaustively verified.

References

[1] V. Reshetnikov and the ProveIt research collection, *Analysis/Polylogarithms manuscript*, retrieved 12 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt research collection, *Incoming research archives*, retrieved 12 October 2026. The supplied source manifest identifies all twelve retrieved archives. https://github.com/VladimirReshetnikov/ProveIt/tree/main/docs/incoming.

[3] ProveIt research collection, *Ordered Boundary Identities*, incoming package dated 11 October 2026, especially `sections/08_gamma_basis.tex`. Archive: `ProveIt_Ordered_Boundary_Identities_2026-10-11.zip`, in [2].

[4] ProveIt research collection, *Polygamma Moments and Stieltjes Resonances*, incoming package dated 12 October 2026, especially `sections/02_gap_reconstruction.tex`, `sections/03_multiscale_collisions.tex`, and `sections/05_source_status.tex`. Archive: `ProveIt_Polygamma_Moments_and_Stieltjes_Resonances_2026-10-12.zip`, in [2].

[5] ProveIt research collection, *Gamma Normalizations and Colored Moments*, incoming package dated 12 October 2026. The two-cluster dictionary and its further research questions are the starting point for Section 3. Archive: `ProveIt_Gamma_Normalizations_and_Colored_Moments_2026-10-12.zip`, in [2].

[6] ProveIt research collection, *Universal Gamma Normal Forms*, expanded incoming package dated 11 October 2026, especially the universal coefficient theorem and `sections/11_audit.tex`, questions 1 and 6. Archive: `ProveIt_Universal_Gamma_Normal_Forms_2026-10-11(2).zip`, in [2].

[7] ProveIt research collection, *Bilinear Tails and Harmonic Determinants*, report in the Stieltjes-harmonic-resonance continuation, especially the compact inverse-row question. Source location is recorded in `INTEGRATION_NOTES.txt`; see also the later reformulation in [4].

[8] NIST Digital Library of Mathematical Functions, Chapter 5, especially Sections 5.7, 5.8, 5.11 and 5.15. https://dlmf.nist.gov/5.

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